

3

MEASUREMENT AND SPACE

SURFACE AREA AND VOLUME

Some theme parks have wave pools, which are big swimming pools that simulate the movement of the water at a beach. A large volume of water is quickly released into one end of the pool, which produces a large wave that moves from one end of the pool to the other. The excess water in the pool is recycled so that it can be used to produce more waves.



Chapter outline

		Working mathematically				
3.01	Absolute error and percentage error	U	F		R	C
3.02	Rounding error	U	F		R	C
3.03	Surface areas of prisms and cylinders	U	F	PS	R	C
3.04	Surface area of a pyramid*	U	F	PS	R	
3.05	Surface areas of cones and spheres*	U	F	PS	R	C
3.06	Surface areas of composite solids*	U	F	PS	R	
3.07	Volumes of prisms and cylinders	U	F	PS	R	
3.08	Volumes of pyramids, cones and spheres*	U	F	PS	R	C
3.09	Volumes of composite solids*	U	F	PS	R	
3.10	Areas of similar figures*	U	F		R	C
3.11	Surface areas and volumes of similar solids*	U	F		R	

***PATH**

Wordbank

absolute error The greatest possible error of a measurement, equal to half of the precision (smallest unit) on the measuring instrument's scale.

cross-section A 'slice' of a solid, taken across the solid rather than along it

curved surface area The area of the curved surface of a solid, such as a cylinder or sphere. The curved surface of a cylinder is a rectangle when flattened.

pyramid A solid with a polygon for a base and triangular faces that meet at a point called the apex.

rounding error The difference between a calculated answer and an exact answer due to rounding in measurement and calculations.

sector A region of a circle cut off by 2 radii, shaped like a piece of pizza

slant height The height of a pyramid or cone from its apex (top) to its base along a side face rather than its perpendicular height.

surface area The total area of all faces of a solid shape

Quiz
Wordbank 3

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (13):

3.03 Surface area of a prism
- Surface areas of prisms
- Surface area of a cylinder

3.04 Surface area of a pyramid

3.05 Surface area of a cone and sphere
- Surface area and volume of a cone

3.07 Capacity of a cylinder
- Volumes of prisms and cylinders

3.08 Volume of a pyramid
- Surface area and volume of a cone
- Volume of a sphere

3.10 Areas of similar figures

3.11 Surface areas and volumes of similar solids

Twig videos (6):

3.01 Decimal places: Photo finish

3.06 The power of the Sun

3.07 Bees and their hives
- The incredible strength of ants
- The Menger sponge

3.08 The Pacific Flyer

3.11 The incredible strength of ants

Quizzes (5):

- Wordbank 3
- SkillCheck 3
- Mental skills 3
- Language of maths 3
- Test yourself 3

Skillsheets (2):

SkillCheck Solid shapes • What is volume?

Worksheets (13):

SkillCheck A page of circular shapes

3.01 Accuracy in measurement 2

3.03 Surface area 2 • Nets of solids

3.06 A page of prisms and cylinders

3.07 A page of prisms and cylinders • Volumes of solids • Back-to-front problems • Volume and capacity • Biggest volume

3.08 Back-to-front problems (Advanced)

3.09 Volume and capacity

3.10 Investigating paper sizes

3.11 Areas and volumes of similar figures

Mind map Surface area and volume (Advanced)

Puzzles (4):

SkillCheck Area

3.03 Surface area

3.07 Formula matching game

Language of maths Surface area and volume crossword (Advanced)

Technology (3):

3.07 Volume of a box • Approximating the volume of a cone

3.10 Areas of similar shapes

Spreadsheets (2):

3.07 Volume of a box

3.10 Areas of similar shapes

Presentation (1):

3.09 Volumes of shapes

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

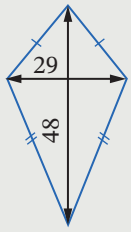
In this chapter you will:

- calculate the absolute error and percentage error of a measurement
- identify and investigate rounding error
- calculate the surface areas, volumes and capacities of right prisms, cylinders and composite solids
- (PATH) calculate the surface areas, volumes and capacities of pyramids, cones, spheres and composite solids
- (PATH) investigate ratios of areas of similar figures
- (PATH) investigate ratios of surface areas and volumes of similar solids

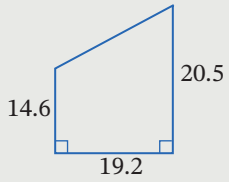
SkillCheck ANSWERS ON P. 615

1 Calculate the area of each shape. All measurements are in centimetres.

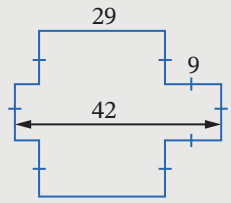
a



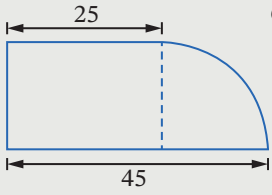
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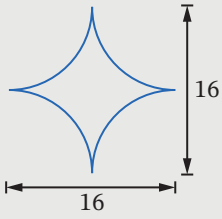
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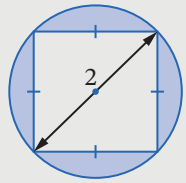
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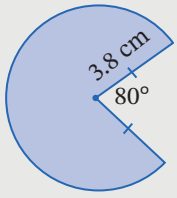


f

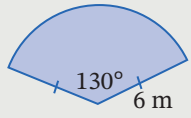


2 Find, correct to one decimal place, the area of each sector.

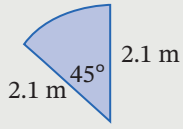
a



b



c



Quiz
SkillCheck 3

Skillsheets
Solid shapes
What is volume?

Puzzle
Area

Worksheet
A page of circular shapes

Absolute error and percentage error 3.01

All measurements are approximate. Whether we are using a ruler to measure length, weighing scales for recipe ingredients or a thermometer to measure body temperature, there is some error in every measurement we make.

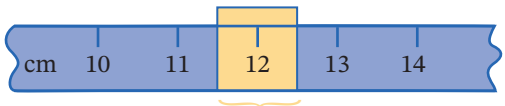
Video
Decimal places: Photo finish

Absolute error of a measurement

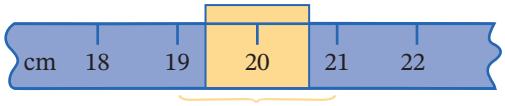
The **absolute error** of a measurement is 0.5 **(half) of the precision** of the measuring instrument's scale.

Worksheet
Accuracy in measurement 2

For example, this ruler has a precision of 1 cm, that is, measurements using this ruler are only accurate to the nearest centimetre.



A measurement of 20 cm on this ruler could actually be any size between 19.5 cm and 20.5 cm.



Strictly speaking, it could not be 20.5 cm because that would round up to 21 cm. The range is actually 19.5 cm up to (but not including) 20.5 cm.

Any measurement in the shaded region should be recorded as 20 cm. The measured length is 20 cm, but the actual length is 20 ± 0.5 cm, meaning '20 cm, give or take 0.5 cm' or '19.5 to 20.5 cm'.

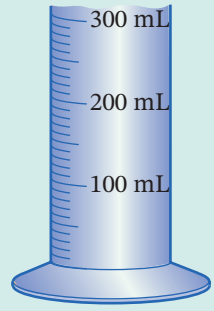
The **absolute error** of this ruler is 0.5 cm, which represents the greatest possible error the actual measurement can differ from the recorded measurement (20 cm), the furthest your measurement could be wrong.

Example 1

Calculate the absolute error of:

a this measuring cylinder

b a measurement of 12.4 cm



Solution

a The cylinder has a precision of 10 mL. — The size of one unit on the scale

Absolute error = 0.5 × 10 mL = 5 mL. — Half of the precision

b A measurement of 12.4 cm has a precision of 0.1 cm. — 12.4 cm is written to the nearest 0.1 cm

Absolute error = 0.5 × 0.1 cm = 0.05 cm.

An absolute error of 1 metre may not be much when measuring the length of a bridge, but it is huge when measuring the length of a bathroom. To understand and compare the sizes of errors when measuring things, we can calculate **percentage error**, the absolute error as a percentage of the measurement.

Percentage error of a measurement

$$\text{Percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100\%$$

The smaller the percentage error, the more accurate the measurement, the closer it is to its actual value.

Example 2

The Channel Tunnel is an underwater railway tunnel connecting England and France. It crosses the English Channel and is 50.45 kilometres long.



Pascal Rossignol/Reuters

Find the percentage error of this measurement, correct to 2 decimal places.

Solution

Precision = 0.01 km

Absolute error = 0.5×0.01 km

$= 0.005$ km

$= \frac{0.005}{50.45} \times 100\%$ $\quad \frac{\text{absolute error}}{\text{measurement}} \times 100\%$

Percentage error = 0.009 91...

$\approx 0.01\ \%$

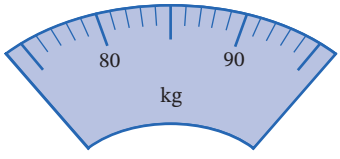
EXERCISE 3.01 ANSWERS ON P. 615

Absolute error and percentage error U F R C

1 For each measuring instrument, find its precision and absolute error. R C

EXAMPLE 1

a



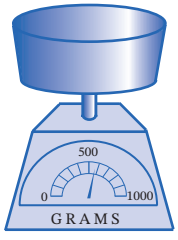
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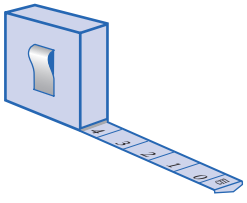
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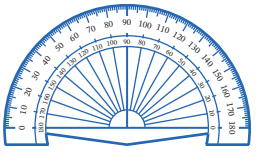
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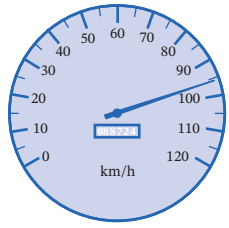


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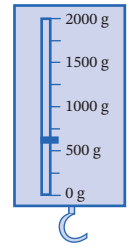


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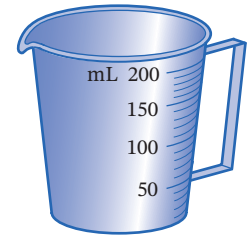
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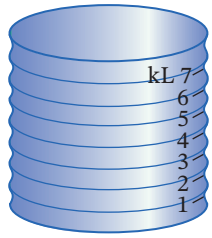
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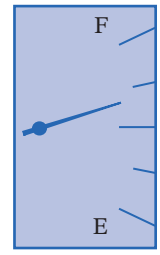
i



j



k



2 Copy and complete this table.

Measurement	Precision	Absolute error
a 58 cm		
b 22.3 mL		
c 31 seconds		
d 175 cm (to the nearest cm)		
e 175 cm (to the nearest 5 cm)		

3 Swara is using a measuring tape marked in millimetres. What is the absolute error of this measurement? Select the correct answer **A**, **B**, **C** or **D**.

A 1 mm **B** 0.5 mm

C 1 cm **D** 0.5 cm

4 Find the absolute error of each measurement.

a 86 mm **b** 2850 L (to the nearest 10 L)

c 9.4 m^2 **d** 700 kg (to the nearest g)

e 700 kg (to the nearest 10 kg) **f** 700 kg (to the nearest 100 kg)

EXAMPLE 2

5 A netball court is 15.25 m wide. Calculate the absolute error and hence find the percentage error of this measurement, correct to 3 decimal places.



Richard Newstead/Moment/Getty Images

6 The world's longest river, the Nile river in Egypt, is 6650 km long.

a If this measurement is made to the nearest 10 km, find its absolute error.

b Hence calculate the percentage error, correct to 4 decimal places.

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7 By comparing percentage errors, determine which measurement is Questions 5 and 6 is more accurate, the width of the netball court or the length of the Nile river. R C

8 A timekeeper measures the time it takes an athlete to complete a race to be 32.08 seconds. Calculate the percentage error of this measurement, correct to 4 decimal places.

9 An Olympic marathon is 42.195 km long. Calculate the percentage error of this measurement, correct to 4 decimal places.

10 Which measurement is Questions 8 and 9 is more accurate, the race time or the marathon length? R C

11 *Sweet Juices* is a company using a new machine to bottle its juice. Bottles of juice have a measured capacity with a percentage error of 0.5% of the 500 mL labelled on each bottle. R C

- **a** What is the absolute error of the machine's scale?
- **b** What is the precision of the machine's scale?
- **c** What is the maximum possible capacity?
- **d** What is the minimum possible capacity?
- **e** Which of the following capacities (in mL) could not have been bottled by the machine?

  500.05, 498.55, 499.10, 503.10, 502.30, 497.25, 498.50, 502.10

12 The length of a piece of wood is measured to be 8 m. The length of a metal rod is measured to be 8.5 m. By using absolute error in your calculations, show that the piece of wood *could* be longer than the metal rod. R C

13 Why is a measurement of 185.0 cm more accurate than a measurement of 185 cm? R C

14 A cube has a measured length of 4 m. Calculate its maximum possible volume and its minimum possible volume. R C

Rounding error 3.02

Another type of error can occur when performing calculations with measurements, where the measured value or partial answer is rounded too early or severely, resulting in a final answer that is inaccurate (not close to the exact answer). This is called **rounding error**.

Consider the area calculations for purchasing carpet for 3 bedrooms.

Room 1 has dimensions 4.2 m by 3.7 m, Room 2 is 3.9 m by 3.4 m, Room 3 is 3.3 m by 3.3 m.

Room 1: $4.2 \times 3.7 = 15.54$ m^2

Room 2: $3.9 \times 3.4 = 13.26$ m^2

Room 3: $3.3 \times 3.3 = 10.89$ m^2

Total area $= 39.69$ m^2



Pixel-Shot/Adobe Stock Photos

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If we were to estimate by rounding measurements to the nearest metre, then:

Room 1: $4 \times 4 = 16$ m^2

Room 2: $4 \times 3 = 12$ m^2

Room 3: $3 \times 3 = 9$ m^2

Total area $= 37$ m^2

Notice that the estimate is too small, and if we had rounded the measurements we would not have bought enough carpet.

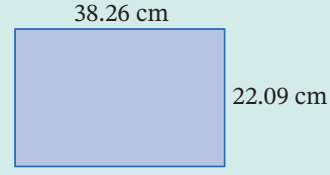
Rounding during calculations can lead to errors in solutions. If approximate values are used in further calculations, the error becomes worse.

The table below shows a number being **rounded** and **truncated** to the nearest whole number, to one decimal place, to 3 decimal places and to 5 decimal places. **Truncate** means to 'cut off' the number at the decimal place regardless of the next digit (or 'rounding down').

10.638 971 564 538 ...	Rounded	Truncated
to the nearest whole number	11	10
to one decimal place	10.6	10.6
to 3 decimal places	10.639	10.638
to 5 decimal places	10.638 97	10.638 97

Example 3

A rectangle has length 38.26 cm and width 22.09 cm.



a Calculate its perimeter and area.

b Round the rectangle's length and width to one decimal place and then calculate its perimeter and area with these values and then find correct to one decimal place the percentage error of the perimeter and area.

c Truncate the rectangle's length and width to the nearest whole number, then calculate its perimeter and area with these values, then find correct to one decimal place the percentage error of the perimeter and area.

Solution

a Perimeter $= 2 \times 38.26 + 2 \times 22.09$

$= 120.7$ cm

Area $= 38.26 \times 22.09$

$= 845.1634$ cm^2

b Length $= 38.3$ cm, width $= 22.1$ cm

Perimeter $= 2 \times 38.3 + 2 \times 22.1$

$= 120.8$ cm

Error $= 120.8 - 120.7$

$= 0.1$ cm

Percentage error $= \frac{0.1}{120.7} \times 100\%$

$= 0.0828 \ldots$

$\approx 0.1\%$

Area $= 38.3 \times 22.1$

$= 846.43$ cm^2

Error $= 846.43 - 845.1634$

$= 1.2666$

Percentage error $= \frac{1.2666}{845.1634} \times 100\%$

$= 0.1498 \ldots$

$\approx 0.1\%$

c Length = 38 cm, width = 22 cm

Perimeter $= 2 \times 38 + 2 \times 22$

$= 120$ cm

Error $= 120.7 - 120$

$= 0.7$ cm

Percentage error $= \frac{0.7}{120.7} \times 100\%$

$= 0.5799 \ldots$

$\approx 0.6\%$

Area $= 38 \times 22$

$= 836$ cm^2

Error $= 845.1634 - 836$

$= 9.1634$

Percentage error $= \frac{9.1634}{845.1634} \times 100\%$

$= 1.0842$

$\approx 1.1\%$

- Notice that the percentage error is higher for area than it is for perimeter: this is due to addition being used to calculate perimeter, whereas by using multiplication to calculate areas, the error is being multiplied.
- Also notice that the percentage error is higher the more we round or truncate the measurements or values. When calculating, it is best to use the exact or most accurate measurements to minimise any rounding errors.

EXERCISE 3.02 ANSWERS ON P. 615

Rounding error UFRC

1 Copy and complete this table to round and truncate the value of π: 3.141 592 6 ...

3.141 592 6 ...	Rounded	Truncated
a to the nearest whole number		
b to 2 decimal places		
c to 4 decimal places		
d to 6 decimal places		

2 A rectangle has length 6.49 m and width 3.82 m.

EXAMPLE 3

a Calculate its area.

b Truncate each measurement to one decimal place, then calculate its approximate area.

c Round each measurement to the nearest whole number, then calculate its approximate area.

3 A circle has radius 7.63 cm.

a Calculate its area correct to 2 decimal places.

b Calculate its area correct to 2 decimal places, using $\pi = 3.14$, then find correct to 2 decimal places the percentage error of the calculated area.

c Round the circle's radius to one decimal place, then calculate its area correct to 2 decimal places, then find correct to 2 decimal places the percentage error of the calculated area.

4 A square has length 11.27 m. R C

- **a** Calculate its perimeter and area.
- **b** Truncate the square's length to one decimal place and use this value to calculate its perimeter and area, then find to one decimal place, the percentage error of the calculated perimeter and area.
- **c** Round the square's length to the nearest whole number and use this value to calculate its perimeter and area, then find to one decimal place the percentage error of the perimeter and area.

5 A rectangle has dimensions 8.25 cm by 6.09 cm. R C

- **a** Calculate its perimeter and area.
- **b** Truncate the rectangle's length and width to one decimal place, then calculate its perimeter and area and find to one decimal place the percentage error of the calculated perimeter and area.
- **c** Round the rectangle's length and width to the nearest whole number, then calculate its perimeter and area and find to one decimal place the percentage error of the perimeter and area.

6 A circle has diameter 8 m, measured correct to the nearest metre. R C

- **a** Calculate the circumference of the circle, correct to 4 decimal places.
- **b** Find the absolute error of the diameter.
- **c** Hence, calculate:
  - **i** correct to 4 decimal places, the smallest possible circumference of the circle
  - **ii** correct to 2 decimal places, the percentage error of the smallest possible circumference
  - **iii** correct to 4 decimal places, the largest possible circumference
  - **iv** correct to 2 decimal places, the percentage error of the largest possible circumference

7 A right-angled triangle has a base length of 60 mm and a perpendicular height of 90 mm, measured to the nearest mm. R C

- **a** Calculate the length of the hypotenuse, correct to 4 decimal places.
- **b** Find the absolute error of the base length and perpendicular height.
- **c** Hence calculate correct to 4 decimal places the minimum possible length of its hypotenuse and correct to one decimal place its percentage error.
- **d** Calculate the area of this triangle.
- **e** Hence calculate the maximum possible area of the triangle and, correct to one decimal place, its percentage error.

8 A rectangular swimming pool has side lengths of 3 m and 8 m, correct to the nearest metre. A fence needs to be placed 1 m out from the edge of the pool enclosing the entire pool. Calculate the maximum and minimum perimeter of this fence. R

Surface areas of prisms and cylinders 3.03

Surface area

The **surface area** of a solid is the total area of all the faces of the solid. To calculate the surface area of a solid, find the area of each face and add the areas together.

For the surface area of an open solid, we only count the *external* surfaces, not the internal ones (so that we don't count each surface twice).

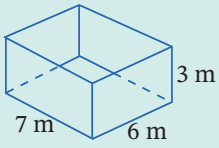
Worksheets
Surface area 2
Nets of solids

Videos
Surface area of a prism
Surface areas of prisms

Example 4

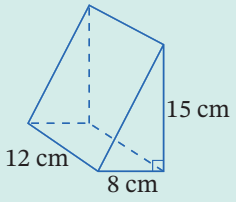
Find the surface area of each prism.

a



Open rectangular prism

b



Closed triangular prism

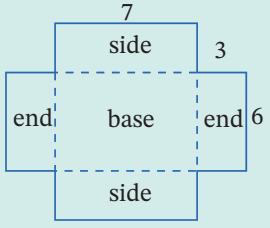
Solution

a This open prism has 5 faces.

Surface area = 2 ends + 2 sides + base

$= 2 \times (3 \times 6) + 2 \times (3 \times 7) + (6 \times 7)$

$= 120 \text{ m}^2$



b This prism has 5 faces.

$m^2 = 8^2 + 15^2$

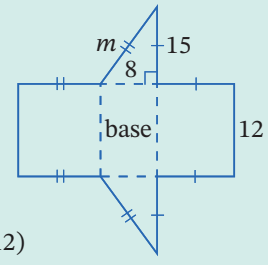
$= 289$

$m = 17$

Surface area = 2 triangles + 3 rectangles

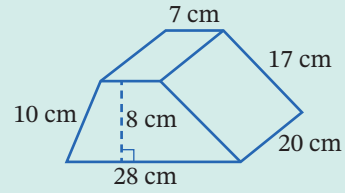
$= 2 \times \left(\frac{1}{2} \times 8 \times 15\right) + (17 \times 12) + (8 \times 12) + (15 \times 12)$

$= 600 \text{ cm}^2$



Example 5

Calculate the surface area of this trapezoidal prism.



Solution

This trapezoidal prism has 6 faces:

2 identical trapeziums (front and back) and 4 different rectangles.

Area of each trapezium $= \frac{1}{2} \times (7 + 28) \times 8$

$= 140 \text{ cm}^2$

Surface area $= (2 \times 140) + (20 \times 7) + (20 \times 10) + (20 \times 28) + (20 \times 17)$

$= 1520 \text{ cm}^2$

Puzzle
Surface area

Video
Surface area of a cylinder

Surface area of a closed cylinder

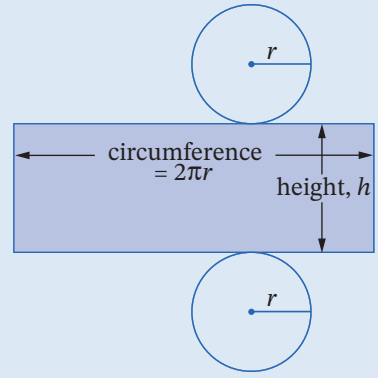
$$SA = 2\pi r^2 + 2\pi rh$$

where r = radius of circular base

h = perpendicular height

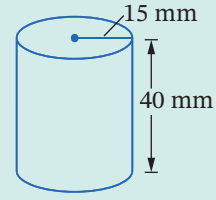
The area of the 2 circular ends $= 2\pi r^2$

and the area of the curved surface $= 2\pi rh$.



Example 6

Find, correct to the nearest mm^2, the surface area of this cylinder.



Solution

Surface area $= 2\pi r^2 + 2\pi rh$

$= 2 \times \pi \times 15^2 + 2 \times \pi \times 15 \times 40$ $\quad r = 15, h = 40$

$= 5183.627...$

$\approx 5184 \text{ mm}^2$

Example 7

Calculate in exact form (in terms of π) the surface area of a cylinder open at one end with radius 12 cm and height 15 cm.

Solution

Surface area = circular end + curved surface

$= \pi \times 12^2 + 2 \times \pi \times 12 \times 15$

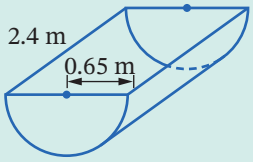
$= 144\pi + 360\pi$

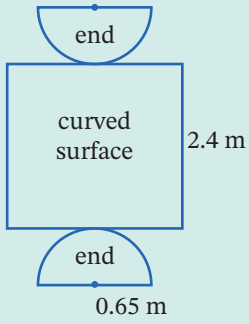
$= 504\pi \text{ cm}^2$

Example 8

Find, correct to 2 **significant figures**, the surface area of an open half-cylinder with radius 0.65 m and length 2.4 m.

Solution





Surface area = 2 semicircle ends + $\frac{1}{2}$ × curved surface

$$= 2 \times \left(\frac{1}{2} \times \pi \times 0.65^2 \right) + \frac{1}{2} \times (2 \times \pi \times 0.65 \times 2.4)$$

$$= 6.2282\ldots$$

$$\approx 6.2 \text{ m}^2$$

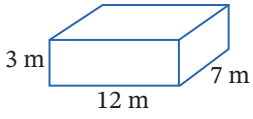
EXERCISE 3.03 ANSWERS ON P. 615

Surface areas of prisms and cylinders U F PS R C

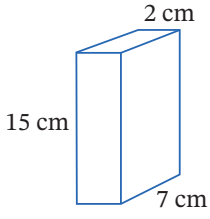
1 Find the surface area of each prism.

EXAMPLE 4

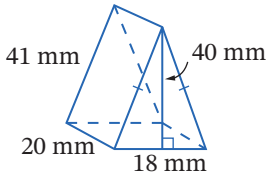
a



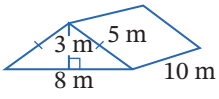
b



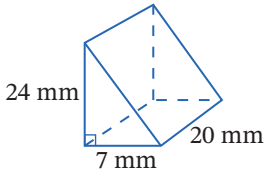
c



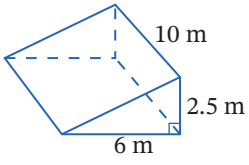
d



e



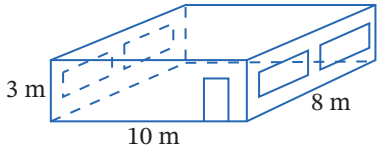
f



2 This classroom is being renovated. Find: PS

a the area of the floor to be carpeted and the total cost, at $55 per square metre.

b the ceiling and wall area to be painted, if the room contains 4 windows, each 2.5 m by 1.5 m, and a doorway 2 m by 0.8 m.

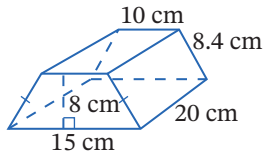


Foundation | Standard | Complex

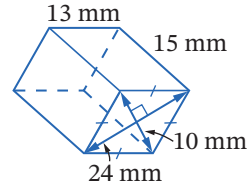
EXAMPLE 5

3 Calculate the surface area of each prism.

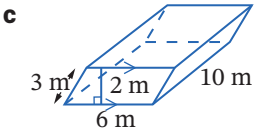
a



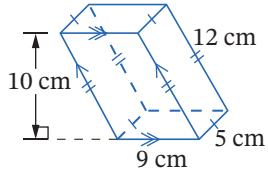
b



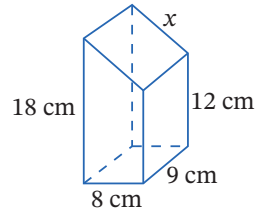
c



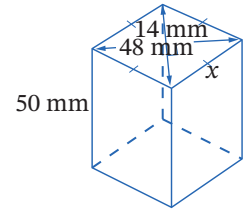
d



e



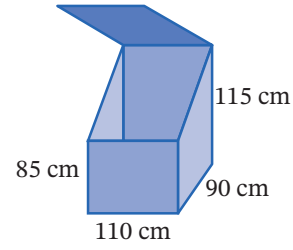
f



4 The wooden toy box is in the shape of a trapezoidal prism.

a Calculate how much timber is required to make the toy box, correct to the nearest cm^2.

b If the price of the timber is $\$25/m^2$, calculate the cost of making the box.



EXAMPLE 6

5 Calculate, correct to one decimal place, the surface area of a cylinder with:

a radius 1.4 m, height 2.2 m

b diameter 45 cm, height 65 cm

c diameter 9 cm, height 24 cm

d radius 1.3 m, height 3.8 m

6 Find, correct to the nearest whole number, the curved surface area of a cylinder with:

a radius 1.5 m, height 3.75 m

b diameter 27 cm, height 41 cm

EXAMPLE 7

7 Calculate, in exact form, the surface area of: R C

a a closed cylinder, radius 6 cm, height 2.5 cm

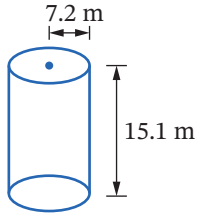
b a cylinder open at one end with radius 1.25 m and height 3.5 m

c a piece of cylindrical tubing, open at both ends, with radius 1.5 cm and length 6.5 m

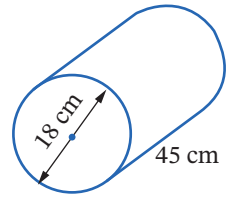
EXAMPLE 8

8 Calculate, correct to the nearest whole number, the surface area of each solid. R

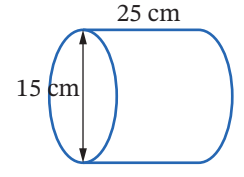
a closed cylinder



b closed cylinder



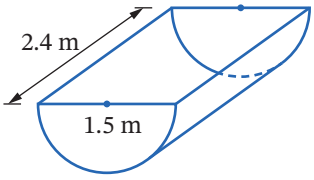
c cylinder with one open end



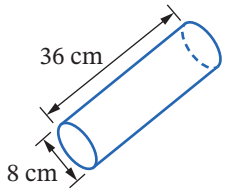
Foundation Standard Complex

9780170479660

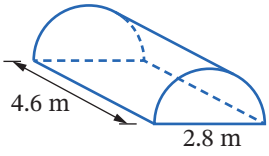
d half cylinder with open top



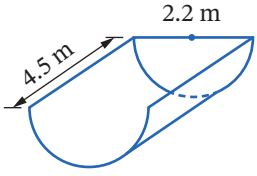
e cylinder open both ends



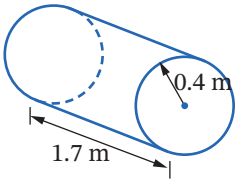
f closed half cylinder



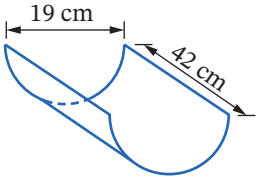
g half cylinder with open top, one end open



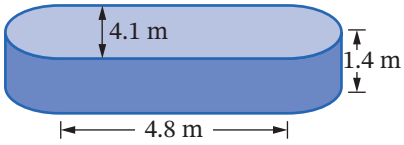
h closed cylinder



i half cylinder, open both ends



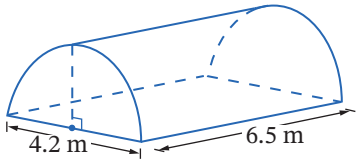
9 The inside of the swimming pool, including the floor, is to be repainted. Find: PS R



a the area to be repainted, correct to one decimal place

b the number of whole litres of paint needed if coverage is 9 m^2 per litre.

10 The diagram shows a tent to be made in the shape of half a cylinder. Find: PS R



a the area of the floor of the tent

b the surface area of the tent, excluding the floor

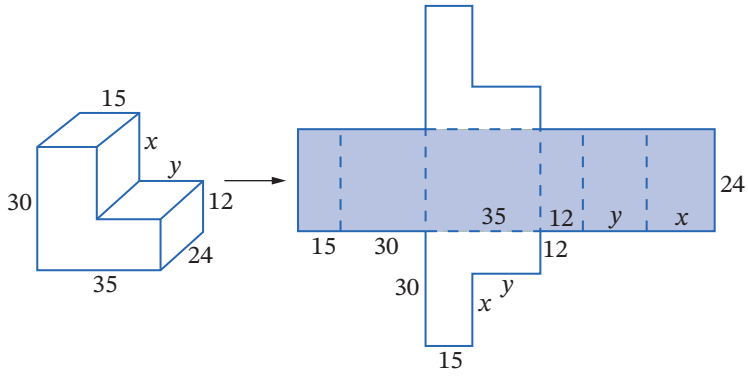
c the total cost of materials for the tent if the material for the flooring costs \$18.50/m^2 and the canvas for the tent costs \$21.75/m^2.

Foundation Standard Complex

Investigation

A surface area shortcut

1 Consider this L-shaped prism and its net. We will find its surface area.



a Find x and y.

b This prism has 8 faces: 2 'L-shaped' ends and 6 rectangles. Instead of calculating the areas of the 6 rectangles separately, we can combine them into one long rectangle, as shaded in the net above. The length of the rectangle is the same as the perimeter of the L-shape. What is the length of this long rectangle?

c What is the area of this long rectangle?

d Copy and complete:

Length of shaded rectangle = p _____ of the L-shape.

e Find the surface area of the prism by copying and completing the following:

Surface area = 2 'L-shaped' ends + shaded rectangle

$= 2 \times (15 \times 30 + 20 \times 12) +$ _____

$=$ _____

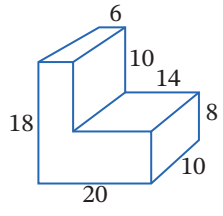
2 From question **1**, it can be seen that the surface area of any prism with end faces of area A and perpendicular height (distance between end faces) h can be calculated using the formula:

$$SA = 2A + Ph$$

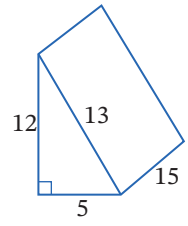
where P = perimeter of the end face.

Use this method to calculate the surface area of each prism. All measurements are in centimetres.

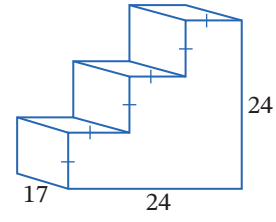
a



b



c



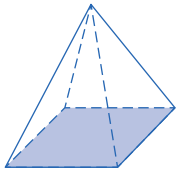
 9780170479660

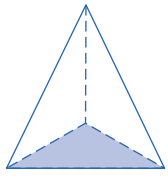
Surface area of a pyramid

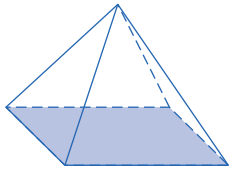
3.04

PATH

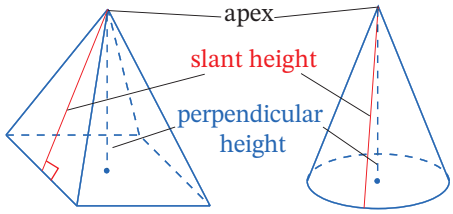
A **pyramid** is a solid shape with a polygon for its base and triangular faces that meet at a point or vertex called its **apex**. Like a prism, a pyramid is named by the shape of its base.


Square pyramid


Triangular pyramid


Rectangular pyramid

A **cone** is a solid shape with a circular base and a curved surface that also has an **apex**. However, a cone is not a pyramid because its base is not a polygon (a circle does not have straight sides).



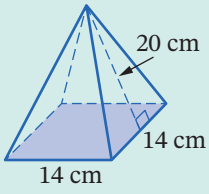
The **slant height** of a pyramid or cone is the height from the apex to the base, along a side face. It is different from the **perpendicular height** of a pyramid or cone, which is the perpendicular distance from the apex to the base.

The surface area of a pyramid is calculated by adding the areas of its base and its triangular faces.

Example 9

Find the surface area of each square pyramid.

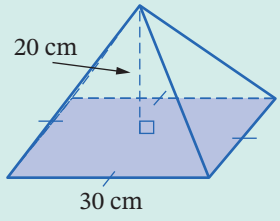
a



This pyramid has a slant height of 20 cm

b

This pyramid has a perpendicular height of 20 cm



Solution

a Surface area = area of square base + area of 4 triangular faces

$$= 14 \times 14 + 4 \times \frac{1}{2} \times 14 \times 20$$

$$= 756 \text{ cm}^2$$

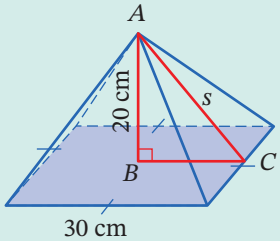
b First find the slant height, s, using Pythagoras' theorem.

In $\triangle ABC$, $s^2 = 20^2 + 15^2$ $\quad BC = \frac{1}{2} \times 30$

$$= 625$$

$$s = \sqrt{625}$$

$$= 25 \text{ cm}$$



Surface area $= 30 \times 30 + 4 \times \frac{1}{2} \times 30 \times 25$

$$= 2400 \text{ cm}^2$$

Video
Surface area of a pyramid

PATH

Example 10

A rectangular pyramid with base 10 cm by 8 cm has a perpendicular height of 15 cm. Find its surface area, correct to one decimal place.

Solution

First find the slant heights AP and AQ.

$AP^2 = AX^2 + XP^2$

$= 15^2 + 4^2 \qquad XP = \frac{1}{2} \times 8$

$= 241$

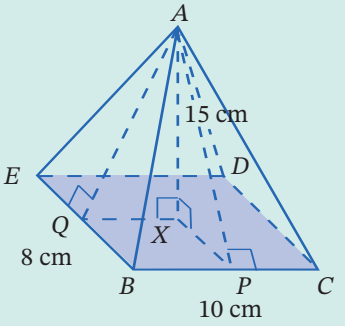
$AP = \sqrt{241}$ cm

$AQ^2 = AX^2 + XQ^2$

$= 15^2 + 5^2 \qquad XQ = \frac{1}{2} \times 10$

$= 250$

$AQ = \sqrt{250}$ cm



It is better to leave the lengths of AP and AQ in surd form rather than round them to decimals so that the final answer is accurate.

Surface area = area of rectangle base + 2 × Area $\triangle ABC$ + 2 × Area $\triangle ADC$

$= 10 \times 8 + 2 \times \frac{1}{2} \times 10 \times \sqrt{241} + 2 \times \frac{1}{2} \times 8 \times \sqrt{250}$

$= 361.7328\ldots$

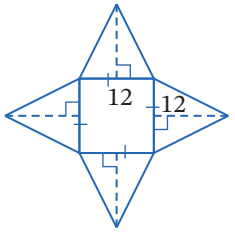
≈ 361.7 cm^2

EXERCISE 3.04 ANSWERS ON P. 615

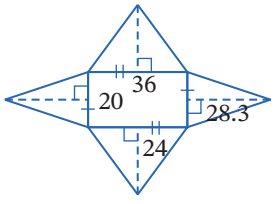
Surface area of a pyramid U F PS R

1 Find the area of each net and hence the surface area of the corresponding pyramid. All measurements are in centimetres.

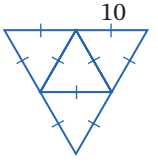
a



b



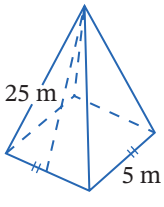
c



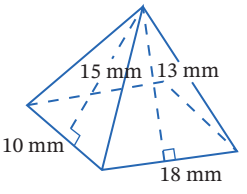
EXAMPLE 9

2 Find the surface area of each pyramid.

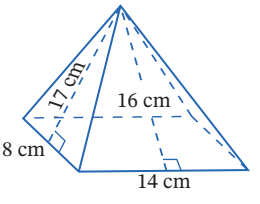
a



b



c

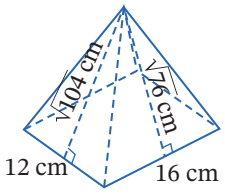


Foundation Standard Complex

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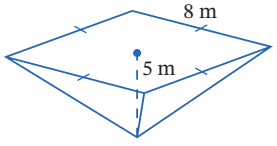
3 Find the surface area of this pyramid. Select the closest answer **A**, **B**, **C** or **D**.

A 288.0 cm²
B 453.9 cm²
C 459.8 cm²
D 471.0 cm²

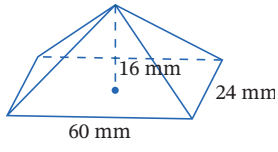


4 Calculate, correct to one decimal place, the surface area of each pyramid. R

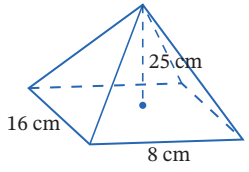
a



b

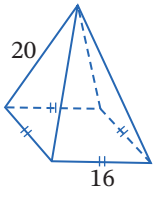


c

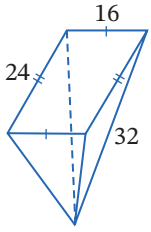


5 Calculate, correct to the nearest square centimetre, the surface area of each pyramid. All measurements are in centimetres. PS R

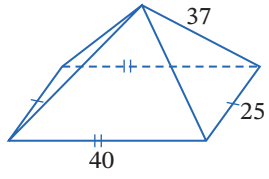
a



b



c



6 The great pyramid of Khufu (or Cheops) in Egypt has a height of 147 m, and each side of its square base measures 230 m. Find its surface area (excluding the base), correct to the nearest square metre. PS R

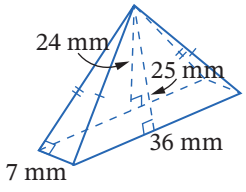
7 The Louvre Art Museum in Paris, France, has a large pyramid at its entrance (built in 1989). It reaches a height of 21.6, has a square base of length 34 m and its sides are covered in rhombus and triangular glass pieces. Calculate the surface area of the glass pyramid (to the nearest m³). PS R



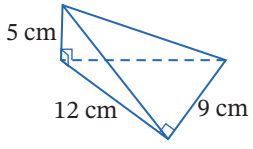
vichie81/Shutterstock.com

8 Calculate, correct to one decimal place, the surface area of each pyramid. PS R

a

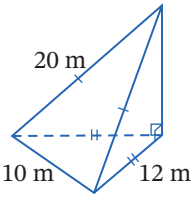


b



All faces are right-angled triangles

c



PATH

9 A square pyramid has a surface area of 4704 m^2 and a base area of 1764 m^2. Find: R

a the length of its base

b the area of each triangular face

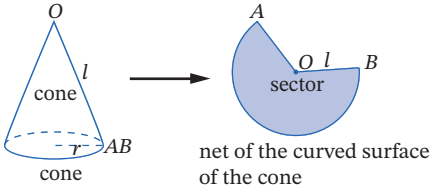
c the slant height of each triangular face

d the perpendicular height of the pyramid

Investigation

The surface area of a cone

The net of a cone is made up of a circle (for the base) and a sector of a circle (for the curved surface). The second diagram below shows the curved surface of a cone.



net of the curved surface of the cone

We can use this fact to find a formula for the curved surface area of a cone. Suppose the cone has base radius r and slant height l. Then the sector must have radius l.

Looking at the sector, the major arc length AB is a fraction of the circumference of the circle and the area of the sector is a fraction of the area of the circle. They should be the same fraction, so:

$$\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\text{Arc length}}{\text{Circumference}}$$

1 The major arc length AB is the same as the circumference AB of the base of the cone in the first diagram. Write an algebraic expression for the circumference of the circle in the first diagram.

2 Write an algebraic expression for the circumference of the complete circle in the second diagram.

3 Write an algebraic expression for the *area* of the complete circle in the second diagram.

4 $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\text{Arc length}}{\text{Circumference}}$ becomes $\frac{\text{Area of sector}}{\pi l^2} = \frac{2\pi r}{2\pi l}$

Copy and complete: $\therefore$ Area of sector $= \frac{2\pi r}{2\pi l} \times \pi l^2 =$ _____

5 But the area of the sector is equal to the curved surface area of the cone.

Copy and complete:

Surface area of cone = area of circular base + area of curved surface

= _____ + _____

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Surface areas of cones and spheres 3.05

PATH

3.05

Surface area of a cone

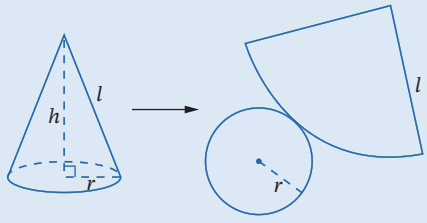
$$SA = \pi r^2 + \pi rl$$

where r = radius of circular base

l = slant height

The area of the circular base = πr^2

and the area of the curved surface = πrl

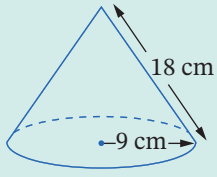


Example 11

For this cone, find correct to one decimal place:

a the curved surface area

b the total surface area



Solution

$r = 9$ cm and $l = 18$ cm

a Curved surface area = πrl

$= \pi \times 9 \times 18$

$= 508.9380\ldots$

≈ 508.9 cm^2

b Total surface area = $\pi r^2 + \pi rl$

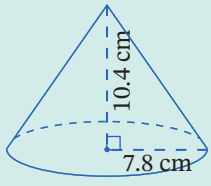
$= \pi \times 9^2 + \pi \times 9 \times 18$

$= 763.4070\ldots$

≈ 763.4 cm^2

Example 12

Find, correct to 2 decimal places, the surface area of this cone.



Solution

First calculate the slant height:

$l^2 = 7.8^2 + 10.4^2$

$= 169$

$l = \sqrt{169}$

$= 13$

Surface area = $\pi r^2 + \pi rl$

$= \pi \times 7.8^2 + \pi \times 7.8 \times 13$

$= 509.6919\ldots$

≈ 509.69 cm^2

Videos
Surface area of a cone and sphere

Surface area and volume of a cone

A **sphere** is a ball shape, completely round. A **hemisphere** is half a sphere.

Surface area of a sphere

$SA = 4\pi r^2$

where r = radius of the sphere

Note that the surface area of a sphere is 4 times the area of a circle that shares its centre.

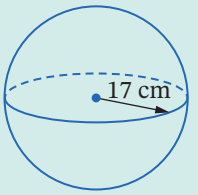


PATH

Video
Surface area of a cone and sphere

Example 13

Find, correct to 2 decimal places, the surface area of this sphere.



Solution

Surface area $= 4\pi r^2$

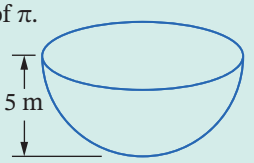
$= 4 \times \pi \times 17^2$

$= 3631.6811...$

$= 3631.68 \text{ cm}^2$

Example 14

Find the surface area of this closed hemisphere in exact form, in terms of π.



Solution

Surface area = Area of circular base + Curved surface area

$= \pi r^2 + \frac{1}{2} \times 4\pi r^2$

$= \pi r^2 + 2\pi r^2$

$= 3\pi r^2$

$= 3 \times \pi \times 5^2$

$= 75\pi \text{ m}^2$

Answers written as surds or in terms of π are exact because they are not decimal approximations.

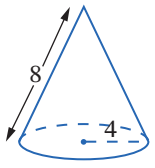
EXERCISE 3.05 ANSWERS ON P. 615

Surface areas of cones and spheres U F PS R C

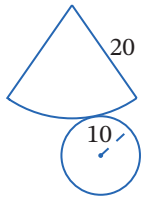
EXAMPLE 11

1 Calculate, correct to the nearest cm^2, the curved surface area of each cone. All measurements are in centimetres.

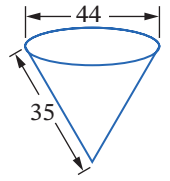
a



b

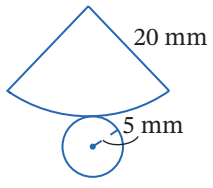


c

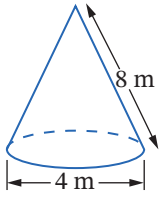


2 Find, correct to one decimal place, the total surface area of each cone.

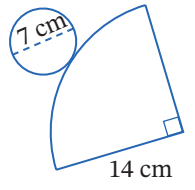
a



b



c



Foundation | Standard | Complex

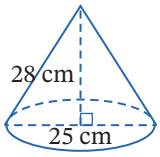
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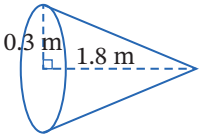
3.05

3 Calculate, correct to one decimal place, the surface area of each cone. R

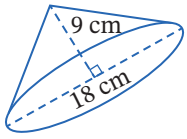
a



b

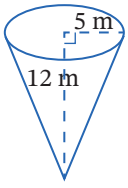


c

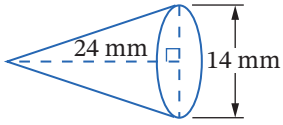


4 Find in exact form (in terms of π) the total surface area of each cone. R C

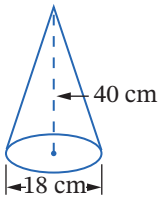
a



b



c

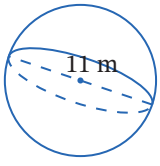


5 Calculate, correct to one decimal place, the surface area of each sphere.

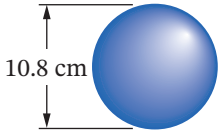
a



b

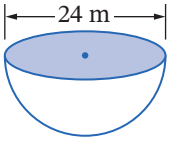


c

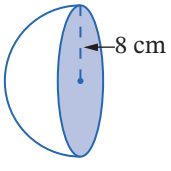


6 Find in exact form the surface area of each closed hemisphere. R C

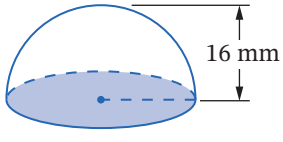
a



b



c



7 Calculate the surface area of a closed hemisphere with radius 1.4 m.
Select the closest answer **A**, **B**, **C** or **D**. R

A 12.32 m^2 **B** 18.47 m^2 **C** 24.63 m^2 **D** 49.26 m^2

8 Find, correct to the nearest square metre, the surface area of: PS R

a a sphere with diameter 10 m

b an open cone with base radius 10 m, slant height 20 m

c an open hemisphere with radius 10 m

d a cone with base diameter 10 m, perpendicular height 20 m

9 The Earth has a radius of approximately 6378 km. PS R C

a Calculate the surface area of the Earth in scientific notation, correct to 3 significant figures.

b If 70.9% of the Earth's surface is covered by water, calculate the land area of the Earth's surface, correct to 3 significant figures.

c The land area of Australia is 7 692 000 km^2. Calculate Australia's land area as a percentage of the Earth's land area (correct to 2 decimal places).

EXAMPLE 12

EXAMPLE 13

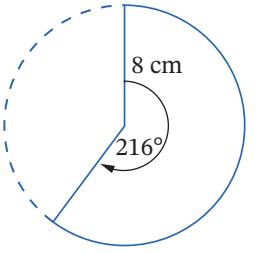
EXAMPLE 14

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10 Find the amount of sheet metal needed to form a conical funnel of base radius 30 cm and vertical height 50 cm, allowing for a 0.5 cm overlap at the join. PS R C

11 A cone is made from a sector of a circle with radius 8 cm and central angle 216°. Find: PS R C



a the length of the arc of the circle which forms the circumference of the cone's base

b the radius of the cone's base

c the slant height of the cone

d the total surface area of the cone, including the base

12 Find the radius of each solid if it has a surface area of 6000 mm^2. Give your answer correct to 3 significant figures. R

a a sphere **b** a closed hemisphere **c** an open hemisphere

13 A cone has a surface area of 2000 cm^2. If the area of its base is 150 cm^2, find: R

a the radius of its base **b** its slant height **c** its perpendicular height

Quiz
Mental skills 3

Mental skills 3: Maths without calculators ANSWERS ON P. 616

Time differences

1 Study each example.

a What is the time difference between 11:40 a.m. and 6:15 p.m.?

From 11:40 a.m. to 5:40 p.m. = 6 hours

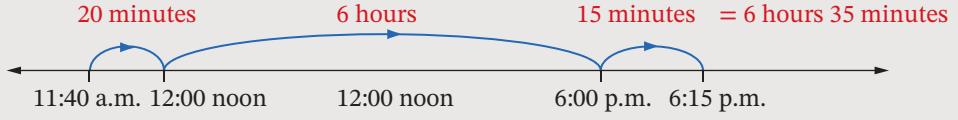
Count: '11:40, 12:40, 1:40, 2:40, 3:40, 4:40, 5:40'

From 5:40 a.m. to 6:00 p.m. = 20 min

From 6:00 p.m. to 6:15 p.m. = 15 min

5 hours + 20 min + 15 min = 6 hours 35 min

OR:



b What is the time difference between 20:30 and 01:20?

From 20:30 to 00:30 = 4 hours (24 − 20 = 4)

From 00:30 to 01:00 = 30 min

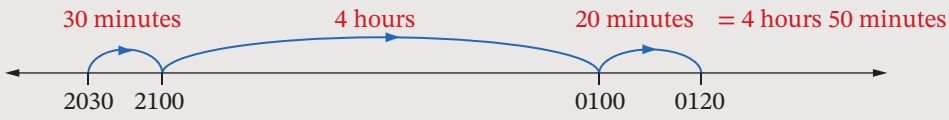
From 01:00 to 01:20 = 20 min

4 hours + 30 minutes + 20 minutes = 4 hours 50 minutes

Foundation Standard Complex

9780170479660

OR:



2 Now find the time difference between:

a 11:10 a.m. and 7:40 p.m.
b 6:20 p.m. and 12:00 midnight
c 4:45 p.m. and 8:10 p.m.
d 2:35 a.m. and 10:50 a.m.
e 1:05 p.m. and 12:30 a.m.
f 9:35 a.m. and 11:15 a.m.
g 04:25 and 09:35
h 14:40 and 20:25
i 7:55 a.m. and 3:50 p.m.
j 2:40 p.m. and 10:20 p.m.

Surface areas of composite solids 3.06

When calculating surface areas of composite solids, remember **not** to include the areas common to both solids.

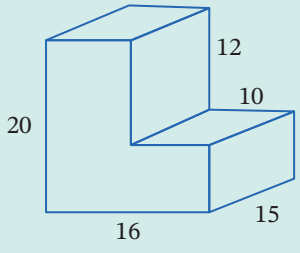
Video
The power of the Sun

Worksheet
A page of prisms and cylinders

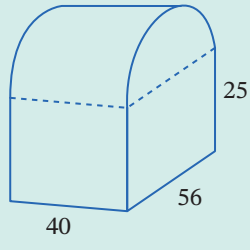
Example 15

Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres.

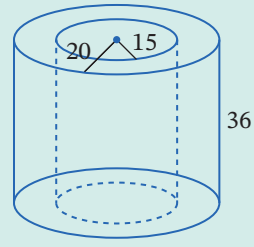
a



b

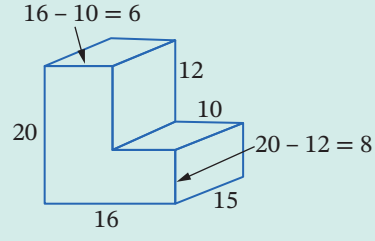


c



Solution

a This prism has 8 faces: 2 identical L-shapes (front and back) and 6 different rectangles.



Area of L-shape $= 16 \times 20 - 10 \times 12$
$= 200 \text{ cm}^2$

Surface area $= (2 \times 200) + (6 \times 15) + (12 \times 15) + (10 \times 15) + (8 \times 15) + (16 \times 15) + (20 \times 15)$
$= 1480 \text{ cm}^2$

Note that the 6 rectangles can also be thought of as one long rectangle of width 15 cm.

Length of long rectangle = perimeter of L = 72

Surface area $= (2 \times 200) + (72 \times 15)$

$= 1480 \text{ cm}^2$

b The solid is made up of a half-cylinder (3 faces) and a rectangular prism (5 faces).

Surface area of half-cylinder
= 2 semi-circular ends + curved surface area

$= 2 \times \frac{1}{2} \times \pi \times 28^2 + \frac{1}{2} \times 2 \times \pi \times 28 \times 40$

$= 5981.5924... \text{ cm}^2$

Radius of semi-circle $= \frac{1}{2} \times 56$
$= 28$

Do not round this partial answer, or the final answer will be inaccurate.

Surface area of rectangular prism $= (2 \times 40 \times 25) + (2 \times 56 \times 25) + (40 \times 56)$

$= 7040 \text{ cm}^2$

Total surface area $= 5981.5924... + 7040$

$= 13021.5924...$

$\approx 13021.6 \text{ cm}^2$

c The hollow cylinder is made up of 2 annulus (ring) faces, an outside curved surface area and an inside curved surface area.

Surface area of annulus faces $= 2 \times (\pi \times 20^2 - \pi \times 15^2)$

$2 \times$ area between 2 circles

$= 1099.5574... \text{ cm}^2$

Outside curved surface area $= 2 \times \pi \times 20 \times 36$

$= 4523.8334... \text{ cm}^2$

Inside curved surface area $= 2 \times \pi \times 15 \times 36$

$= 3392.9200... \text{ cm}^2$

Total surface area $= 1099.5574... + 4523.8934... + 3392.9200...$

$= 9016.3709...$

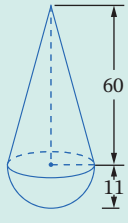
$= 9016.4 \text{ cm}^2$

PATH

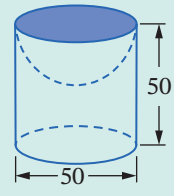
Example 16

Find, correct to the nearest square centimetre, the surface area of each solid. All measurements are in centimetres.

a



b



Solution

PATH

a Surface area = curved surface area of cone + curved surface area of hemisphere

$$= \pi rl + \frac{1}{2} \times 4\pi r^2$$

$r = 11$, $h = 60$ and $l = ?$

$l^2 = 11^2 + 60^2$

$= 3721$

$l = \sqrt{3721}$

$= 61$ cm

Surface area $= \pi \times 11 \times 61 + \frac{1}{2} \times 4 \times \pi \times 11^2$

$= 2868.2740 \ldots$

≈ 2868 cm^2

b Surface area = curved surface of cylinder + circular base + curved surface of hemisphere

$= 2\pi rh + \pi r^2 + \frac{1}{2} \times 4\pi r^2$

$= 2\pi rh + 3\pi r^2$ $\quad \pi r^2 + 2\pi r^2 = 3\pi r^2$

$= 2 \times \pi \times 25 \times 50 + 3 \times \pi \times 25^2$ $\quad r = \frac{1}{2} \times 50 = 25$

$= 13\,744.4678 \ldots$

$\approx 13\,744$ cm^2

3.06

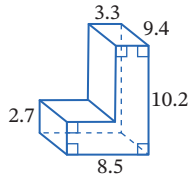
EXERCISE 3.06 ANSWERS ON P. 616

Surface areas of composite solids U F PS R

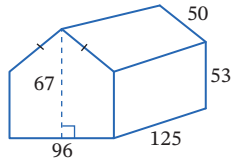
EXAMPLE 15

1 Find the surface area of each solid. All measurements are in centimetres. PS R

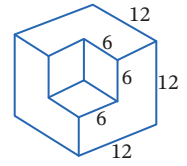
a



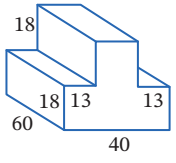
b



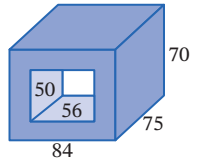
c



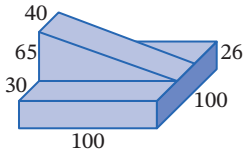
d



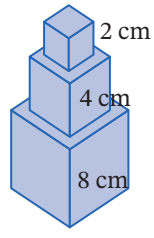
e



f



2 3 cubes of length 2 cm, 4 cm and 8 cm are glued on top of each other. Calculate the surface area of the new solid. PS R

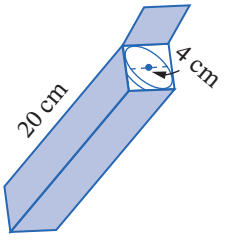


Foundation Standard Complex

3 Circular cracker biscuits of diameter 4 cm are packed in a cardboard box of length 20 cm.

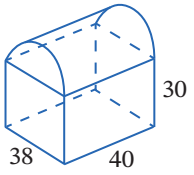
a Calculate the surface area of the box.

b How much cardboard would be saved if the biscuits were packed into a cylindrical box?

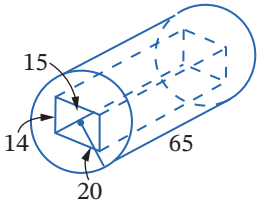


4 Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres. PS R

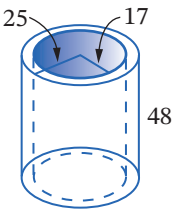
a



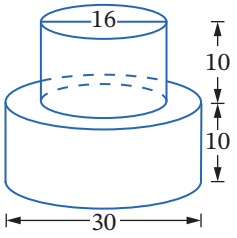
b



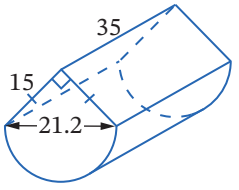
c



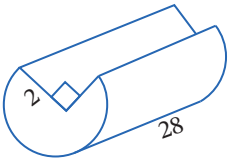
d



e



f

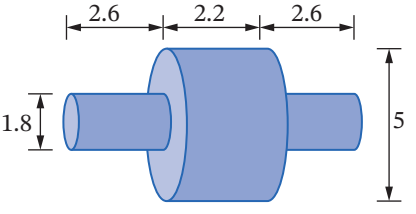


5 Calculate the surface area of the solid, correct to one decimal place. All measurements are in centimetres. R

Select the correct answer **A**, **B**, **C** or **D**.

A 86.0 cm² **B** 103.2 cm²

C 108.3 cm² **D** 113.4 cm²



6 A wedding cake with 3 tiers rests on a table. Each tier is 6 cm high. The layers have radii of 20 cm, 15 cm and 10 cm respectively. Find the total visible surface area, correct to the nearest cm². PS R

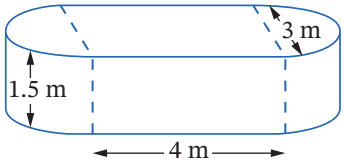


Brian Barney/Shutterstock.com

Foundation Standard Complex

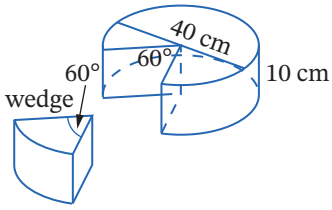
9780170479660

7 **a** Find, correct to 2 decimal places, the total external area of the wall of this above-ground swimming pool.



b Calculate the area of the water surface, correct to the nearest m². PS R

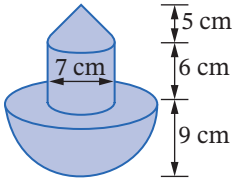
8 A wedge of cheese is cut from a cylindrical block of height 10 cm and diameter 40 cm. Find the total surface area of the wedge, correct to 2 decimal places. PS R



PATH

EXAMPLE 16

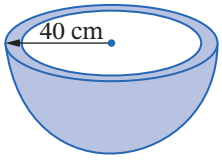
9 Calculate the surface area of the solid. Select **A**, **B**, **C** or **D**. PS R



A 923.98 cm² **B** 962.46 cm²

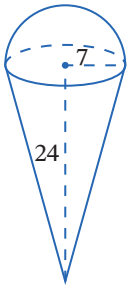
C 736.62 cm² **D** 669.51 cm²

10 Calculate the total surface of a hemispherical bird bath that is 1 cm thick and has an outer radius of 40 cm. PS R

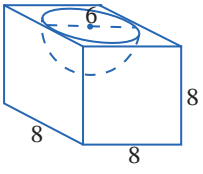


11 Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres. PS R

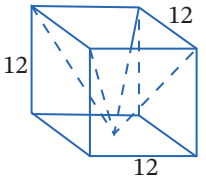
a



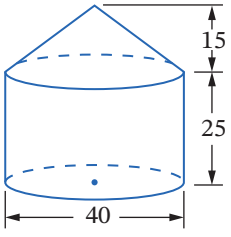
b



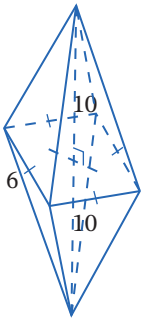
c



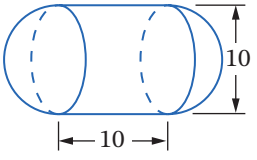
d



e



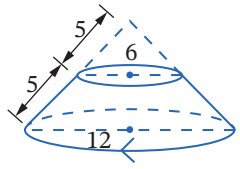
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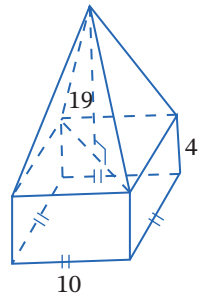
Foundation Standard Complex

PATH

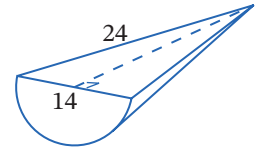
g



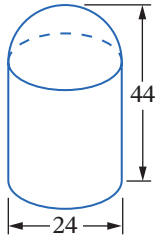
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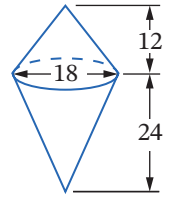
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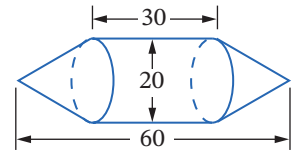
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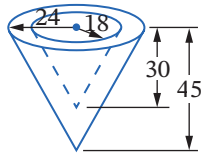
k



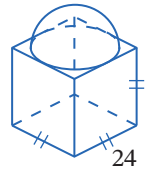
l



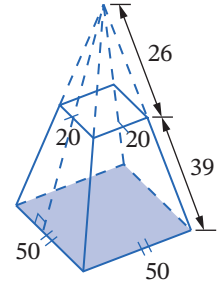
m



n



o



3.07 Volumes of prisms and cylinders

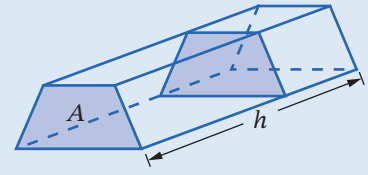
Worksheets
A page of prisms and cylinders
Volumes of solids
Back-to-front problems
Volume and capacity
Biggest volume

Volume of a prism

$V = Ah$

where A = area of base

h = perpendicular height

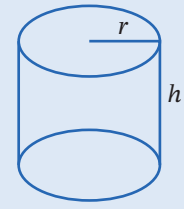


Volume of a cylinder

$V = \pi r^2 h$

where r = radius of circular base

h = perpendicular height



Volume and capacity

1 cm³ contains 1 mL

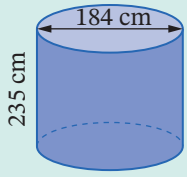
1 m³ contains 1000 L or 1 kL

Example 17

For this cylinder, calculate:

a its volume correct to the nearest cm³

b its capacity in kL, correct to 1 decimal place.



Solution

a Radius $= \frac{1}{2} \times 184$

$= 92$ cm

$V = \pi \times 92^2 \times 235$

$= 6\,248\,753.451...$

$\approx 6\,248\,753$ cm³

b Capacity $= 6\,248\,753$ mL

$= 6\,248\,753 \div 1000 \div 1000$ kL

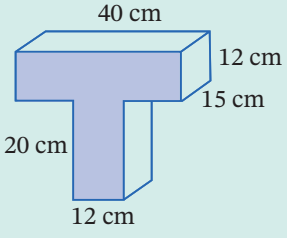
$= 6.248\,753$ kL

≈ 6.2 kL

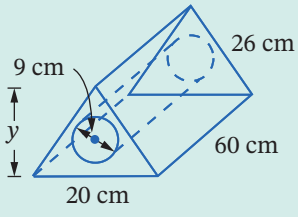
Example 18

Find, correct to the nearest whole number, the volume of each solid.

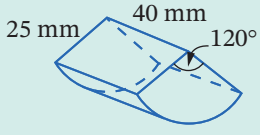
a



b



c



Solution

a $A = 40 \times 12 + 20 \times 12$

$= 720$ cm²

$V = Ah$

$= 720 \times 15$

$= 10\,800$ cm³

b Cross-section is a triangle minus a circle.

$26^2 = y^2 + 10^2$

$y^2 = 26^2 - 10^2$

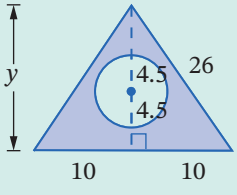
$= 576$

$y = 24$ cm

Radius of circle $= \frac{1}{2} \times 9 = 4.5$

$A = \frac{1}{2} \times 20 \times 24 - \pi \times 4.5^2$

$= 176.3827...$ cm²



$V = Ah$

$= 176.3827... \times 60$

$= 10\,582.9649...$

$\approx 10\,583$ cm³

Technology
Volume of a box

Spreadsheet
Volume of a box

Puzzle
Formula matching game

Videos
Capacity of a cylinder

Volumes of prisms and cylinders

Bees and their hives

The incredible strength of ants

The Menger sponge

3.07

c $A = \frac{120}{360} \times \pi \times 25^2$
$= 654.498...\ \text{mm}^2$

$V = Ah$
$= 654.498... \times 40$
$= 26\,179.938...$
$\approx 26\,180\ \text{mm}^2$

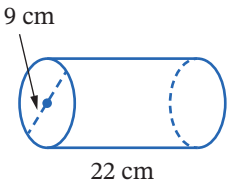
EXERCISE 3.07 ANSWERS ON P. 616

Volumes of prisms and cylinders U F R

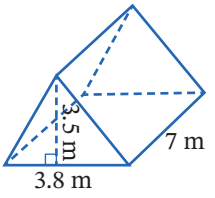
EXAMPLE 17

1 Calculate, correct to one decimal place, the volume of each solid.

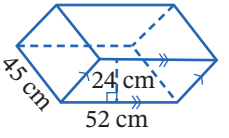
a



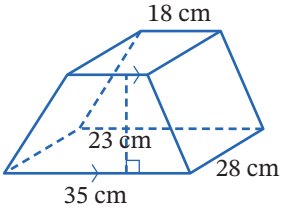
b



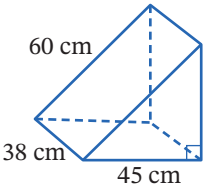
c



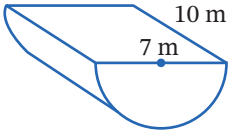
d



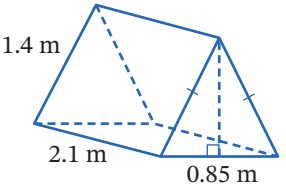
e



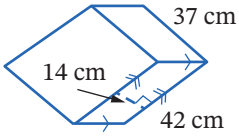
f



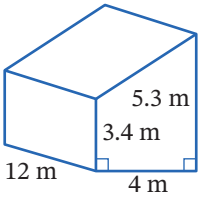
g



h



i

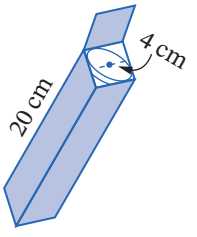


2 Rice crackers of diameter 4 cm are packed in a cardboard box of height 20 cm. Calculate, correct to one decimal place:

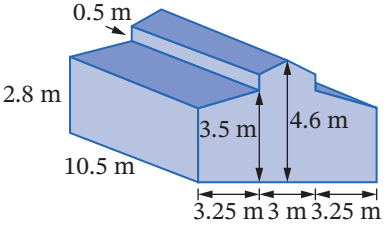
a the volume of the crackers in the box

b the volume of the box

c the percentage of the box that is empty space.



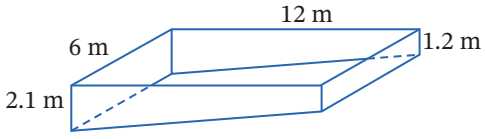
3 Calculate, correct to one decimal place, the volume of the shed. R



Foundation Standard Complex

9780170479660

4 This swimming pool is 12 m long and 6 m wide. The depth of the pool ranges from 1.2 m to 2.1 m.



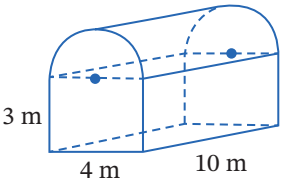
a Calculate the capacity of this pool in litres.

b If the pool is filled so that the water is 8 cm from the top of the pool, calculate the amount of water in the pool (to the nearest litre).

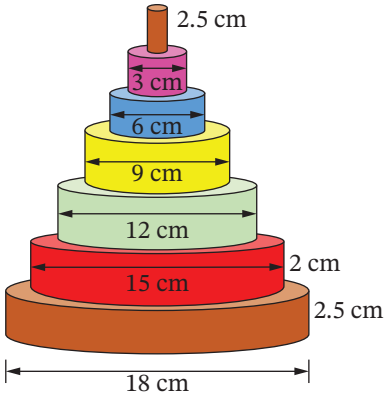
5 An Olympic sized swimming pool is 50 m long, 25 m wide and 2 m deep. What is the capacity of an Olympic pool in litres?

6 **a** Find, correct to 2 decimal places, the volume of this greenhouse.

b If this greenhouse costs 0.5c per m^3 per hour to heat, how much is this per day (correct to the nearest cent)?



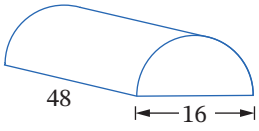
7 A baby's toy involves placing 5 coloured cylinders on a wooden stick (diameter 1 cm) on a base (diameter 18 cm). The cylinders are of varying diameters with a height of 2 cm. Find the volume of the toy, including the base and peg, correct to one decimal place. R



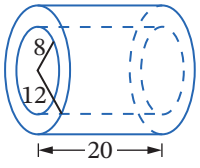
8 Find, correct to 2 decimal places where appropriate, the volume of each solid. All lengths shown are in centimetres. R

EXAMPLE 18

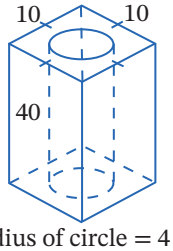
a



b



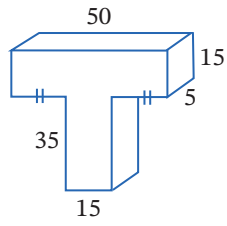
c



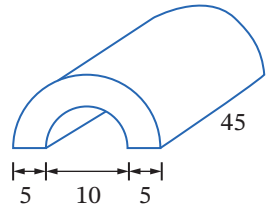
radius of circle = 4 cm

Foundation | Standard | Complex

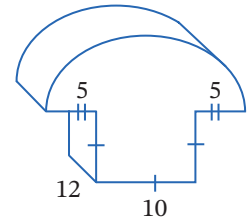
d



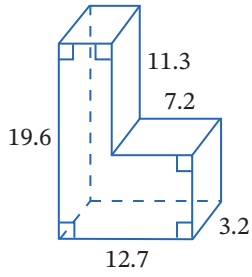
e



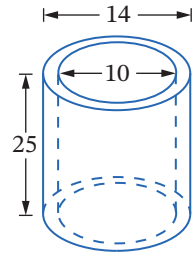
f



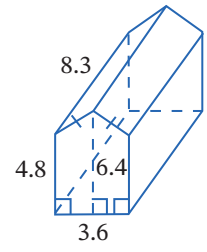
g



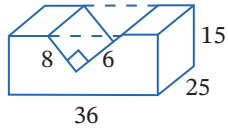
h



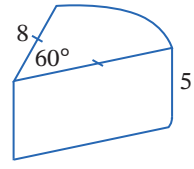
i



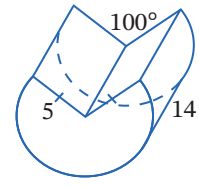
j



k



l



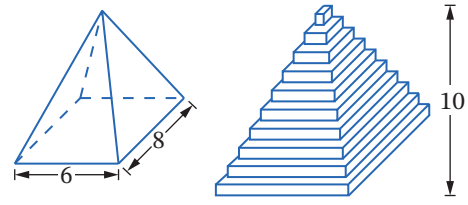
PATH

Technology
Approximating the volume of a cone

Technology

Approximating the volume of a pyramid

In this activity, we will use a spreadsheet to approximate the volume of a rectangular pyramid by slicing it into tiny layers of rectangular prisms of equal thickness.



Let $L = 6$ be the length of the prism, $W = 8$ be the width and $H = 10$ be the height.

The thickness, T, of each layer is given by the formula $T = \dfrac{H}{\text{number of layers}}$.

Starting at the bottom, the length and width of each layer are decreased by the amounts $\dfrac{L}{\text{number of layers}}$ and $\dfrac{W}{\text{number of layers}}$ with each step.

9780170479660

PATH

1 Set up your spreadsheet as shown.

	A	B	C	D	E	F
1						
2			Number of layers =			
3	H	L	W	Thickness of layers	Volume of layer	Sum of volumes
4	10	6	8	=$A$4/$D$2	=B4*C4*D4	=E4
5		=B4-$B$4/$D$2	=$C4-$C$4/$D$2			=E5+F4
6						
7						

2 Let the number of layers be 10. Enter 10 in cell D2.

3 Copy each formula down to row 13.

4 Explain the results in cells E13 and F13.

5 How accurate was your result in F13? Explain.

6 Print out your spreadsheet and paste it into your book.

7 Enter 40 (layers) in cell D2 and copy each formula down to row 43.

8 In 1–2 sentences compare your result in F43 with the previous result in F13 from question 4.

9 Enter each value in cell D2 and copy down the formulas as requested.

a 100 (copy down to row 103)

b 200 (copy down to row 203)

c 400 (copy down to row 403)

10 Use the formula $V = \frac{1}{3}Ah$ to calculate the exact volume of the pyramid.

11 Write a brief report about your results in questions **9** and **10**.

3.08

Volumes of pyramids, cones and spheres 3.08

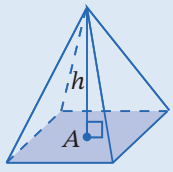
Volume of a pyramid

PATH

$$V = \frac{1}{3}Ah$$

where A = area of the base

h = perpendicular height.

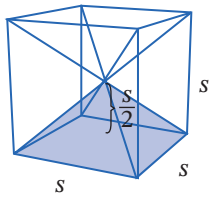


Video
The Pacific Flyer

Worksheet
Back-to-front problems (Advanced)

The volume of a pyramid is $\frac{1}{3}$ of the volume of a prism with the same base and height.

This fact is difficult to prove mathematically, but here's a diagram that might help. This is a square pyramid with dimensions s, s and $\frac{s}{2}$, with an identical pyramid balancing on top of it upside-down.



Now look at the big cube that surrounds these 2 square pyramids.

- Can you see how many such pyramids fit into this big cube exactly?
- What is the volume of the big cube?
- So, what is the volume of one pyramid?
- Is this the same as $V = \frac{1}{3}Ah$?

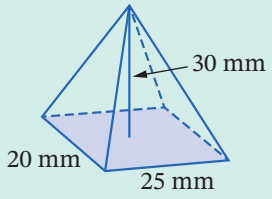
PATH

Video
Volume of a pyramid

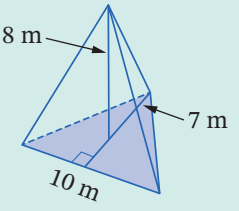
Example 19

Find the volume of each pyramid.

a



b



Solution

a
$$A = 25 \times 20$$
$$= 500 \text{ mm}^2$$
$$V = \frac{1}{3} \times 500 \times 30$$
$$= 5000 \text{ mm}^3$$

b
$$A = \frac{1}{2} \times 10 \times 7$$
$$= 35 \text{ mm}^2$$
$$V = \frac{1}{3} \times 35 \times 8$$
$$= 93\frac{1}{3} \text{ m}^3$$

Example 20

Find the volume of a square pyramid with base length 22 mm and slant height 61 mm.

Solution

First find h, the perpendicular height of the pyramid.

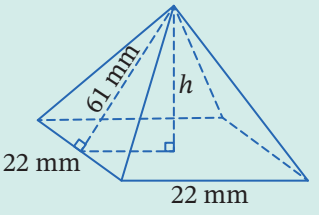
$h^2 = 61^2 - 11^2 = 3600$

$h = 60$ mm

$A = 22^2 = 484$

$V = \frac{1}{3} \times 484 \times 60$

$= 9680 \text{ mm}^3$



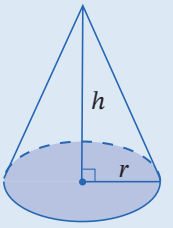
Video
Surface area and volume of a cone

Volume of a cone

$$V = \frac{1}{3}\pi r^2 h$$

where r = radius of the base

h = perpendicular height.

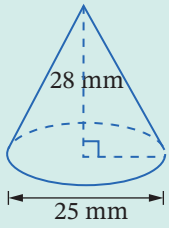


9780170479660

A cone is like a 'circular pyramid' so $V = \frac{1}{3}Ah = \frac{1}{3} \times \pi r^2 \times h = \frac{1}{3}\pi r^2 h$

Example 21

Find, correct to the nearest cubic centimetre, the volume of this cone.



Solution

$V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \pi \times 12.5^2 \times 28 \quad r = \frac{1}{2} \times 25 = 12.5$

$= 4581.4892...$

$\approx 4581 \text{ cm}^3$

Example 22

A cone has a slant edge of 50 mm and a base radius of 15 mm.
Find its volume correct to 2 significant figures.

Solution

$h^2 = 50^2 - 15^2$

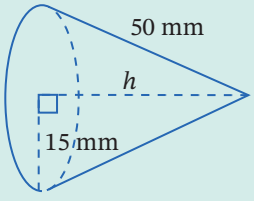
$= 2275$

$h = \sqrt{2275}$

$V = \frac{1}{3} \times \pi \times 15^2 \times \sqrt{2275}$

$= 11\,238.3314...$

$\approx 11\,000 \text{ mm}^3$



Volume of a sphere

$V = \frac{4}{3}\pi r^3$

where r = radius of the sphere.



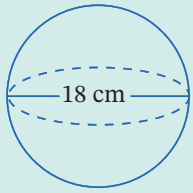
PATH

Video
Volume of a sphere

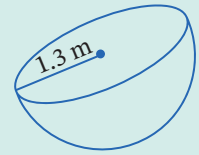
Example 23

Find, correct to 2 significant figures, the volume of each solid.

a



b



Solution

a $V = \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \pi \times 9^3$ $\quad r = \frac{1}{2} \times 18 = 9$

$= 3053.6280...$

$\approx 3100 \text{ cm}^3$

b $V = \frac{1}{2} \times \frac{4}{3}\pi r^3$

$= \frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \pi \times 1.3^3$

$= 4.6013...$

$\approx 4.6 \text{ m}^3$

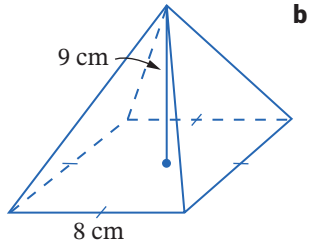
EXERCISE 3.08 ANSWERS ON P. 616

Volumes of pyramids, cones and spheres U F PS R C

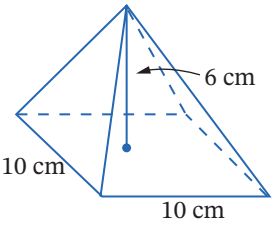
EXAMPLE 19

1 Find, correct to one decimal place, the volume of each pyramid.

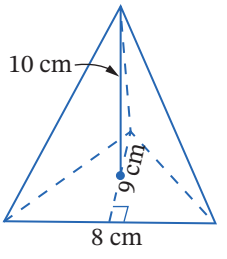
a



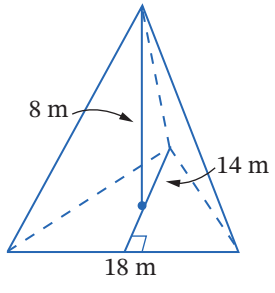
b



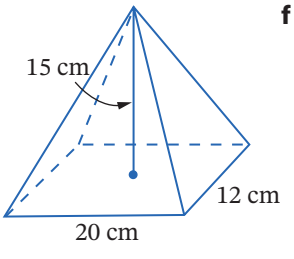
c



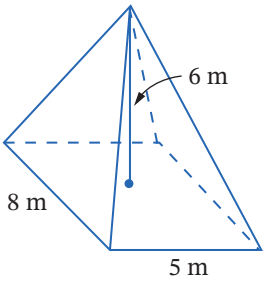
d



e



f



Foundation Standard Complex

9780170479660

PATH

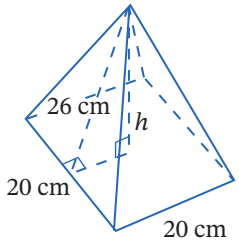
EXAMPLE 20

3.08

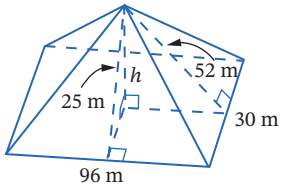
2 For each pyramid, find correct to one decimal place: R

i its perpendicular height **ii** its volume

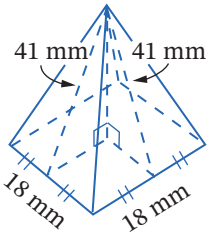
a



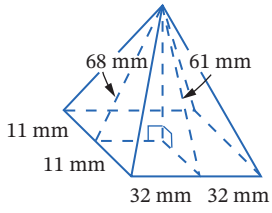
b



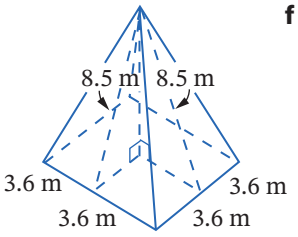
c



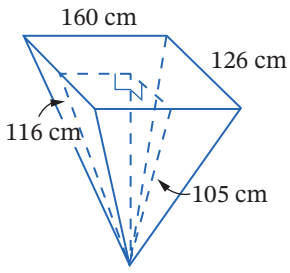
d



e



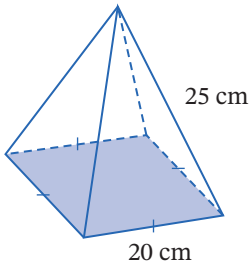
f



3 Calculate the volume of this square pyramid.
Select the closest answer **A**, **B**, **C** or **D**. R

A 3055.1 cm^3 **B** 2000 cm^3

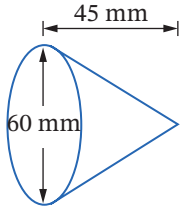
C 3333.3 cm^3 **D** 2748.7 cm^3



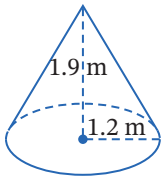
EXAMPLE 21

4 Find, correct to the nearest whole number, the volume of each cone.

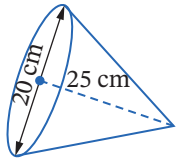
a



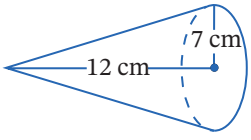
b



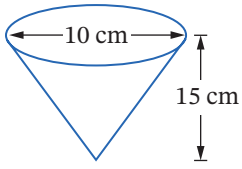
c



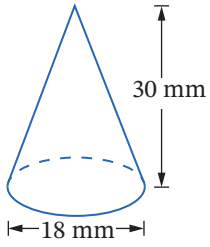
d



e



f



Foundation Standard Complex

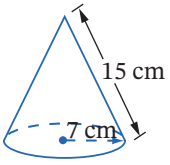
PATH

EXAMPLE 22

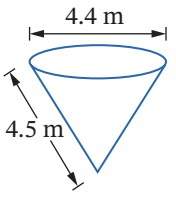
5 For each cone, find correct to one decimal place: R

i its perpendicular height **ii** its volume

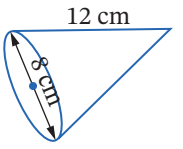
a



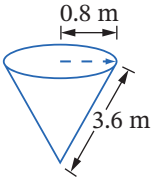
b



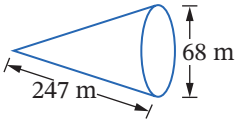
c



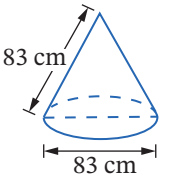
d



e



f



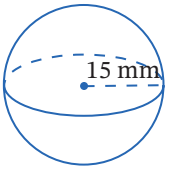
6 Find the volume of a cone with a base diameter of 5.4 cm and a slant height of 9.8 cm. Select **A**, **B**, **C** or **D**. R

A 75 cm³ **B** 72 cm³ **C** 288 cm³ **D** 111.1 cm³

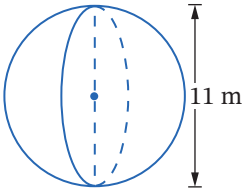
EXAMPLE 23

7 Find the volume of each sphere or hemisphere, correct to one decimal place.

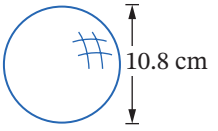
a



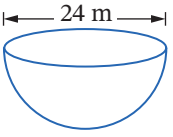
b



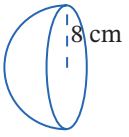
c



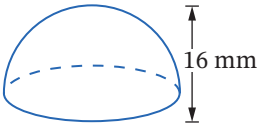
d



e



f

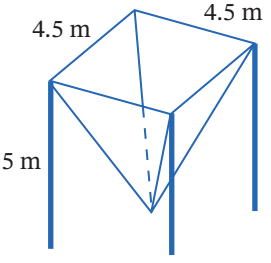


8 The Earth has a radius of approximately 6378 km. Calculate its volume in scientific notation, correct to 3 significant figures. C

9 A grain hopper is in the shape of a square pyramid. PS R C

a Find the volume of grain it holds when full.

b If there are 750 kg of wheat per m³, find the mass of grain in the hopper when it is filled to three-quarters of capacity. Give your answer correct to the nearest tonne.



10 A pyramid has a volume of 360 m³ and a base area of 48 m². Calculate its perpendicular height. R

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11 A square pyramid has a volume of 800 cm³ and a perpendicular height of 12 cm. Calculate, correct to one decimal place, the length of its base. R

12 A cone has a volume of 600 m³ and a base radius of 10 m. Calculate, correct to one decimal place, its perpendicular height. R

13 A cone has a volume of 160 cm³ and a perpendicular height of 20 cm. Calculate, correct to one decimal place, its radius. R

14 The volume of a spherical light is 2664 cm³. Calculate its diameter. Select **A**, **B**, **C** or **D**. R

A 17.2 cm **B** 8.6 cm **C** 17.0 cm **D** 8.5 cm

PATH

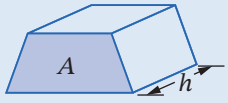
Volumes of composite solids 3.09

Volume formulas

PATH

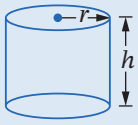
Prism

$V = Ah$



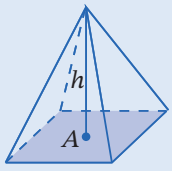
Cylinder

$V = \pi r^2 h$



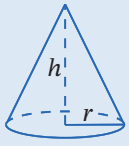
Pyramid

$V = \frac{1}{3}Ah$



Cone

$V = \frac{1}{3}\pi r^2 h$



Sphere

$V = \frac{4}{3}\pi r^3$



Worksheet
Volume and capacity

Presentation
Volumes of shapes

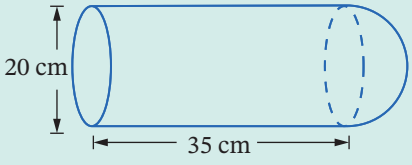
Note that the formulas for **surface area** involve **2 dimensions,** for example, r^2 or rh, while the formulas for **volume** involve **3 dimensions,** for example, r^2h or r^3 or Ah.

Example 24

For this packing capsule, find correct to the nearest whole number:

a its volume

b its capacity in litres



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PATH

Solution

a Volume $= \pi r^2 h + \frac{1}{2} \times \frac{4}{3} \pi r^3$

$= \pi \times 10^2 \times 35 + \frac{2}{3} \times \pi \times 10^3$

$= 13\,089.9693\ldots$

$\approx 13\,090\ \text{cm}^3$

b Capacity $\approx 13\,090$ mL

$= 13.09$ L

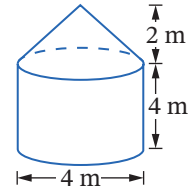
≈ 13 L

EXERCISE 3.09 ANSWERS ON P. 616

Volumes of composite solids U F PS R

EXAMPLE 24

1 This storage tank shown is completely filled with water.

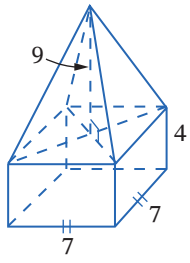


a Calculate to the nearest cubic metre the volume of the tank.

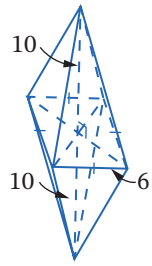
b Find the capacity of the tank, correct to the nearest kilolitre.

2 Find the volume of each solid. All measurements are centimetres. R

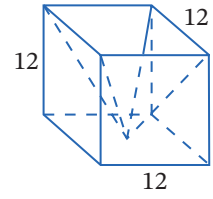
a



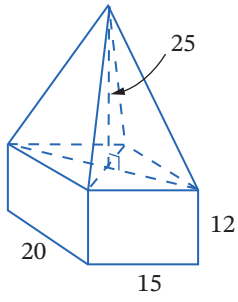
b



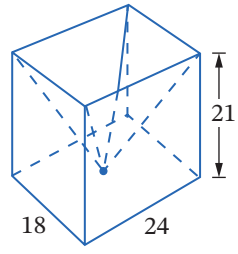
c



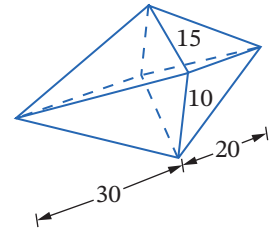
d



e



f



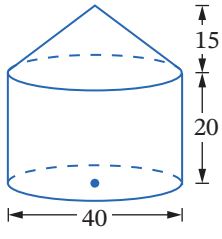
3 For each solid, find:

i the volume (to the nearest cm^3)

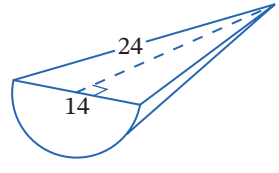
ii the capacity (in litres, correct to 3 decimal places).

All measurements are in centimetres.

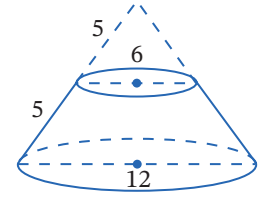
a



b



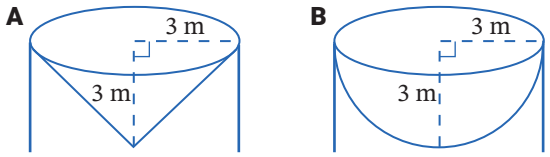
c



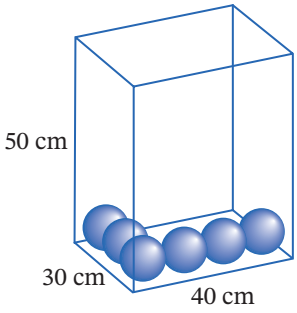
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4 A conical tank (A) and a hemispherical tank (B) have measurements as shown. How much more does tank B hold compared to tank A? Answer correct to 2 decimal places.



5 Spherical balls of diameter 10 cm are stacked inside a box as shown. PS R



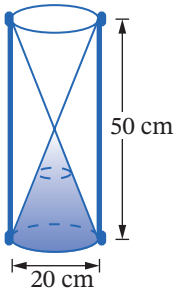
a How many balls will fit in the bottom layer?

b If the balls are stacked in the same manner as the bottom layer until the box is full, how many balls will fit in the box?

c Calculate the volume of the space occupied by the balls when the box is full.

d What percentage of the box is empty space? Give your answer to the nearest whole percentage.

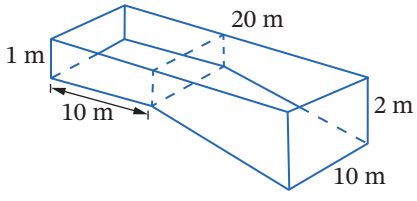
6 The sand in this hourglass takes up $\frac{3}{4}$ of the volume of the bottom cone. PS R



a Calculate, correct to the nearest cubic centimetre, the volume of sand in the hourglass.

b If the sand takes one hour to fall from the top cone to the bottom cone, at what rate is it falling? Give your answer in cm^3/s, correct to 2 significant figures.

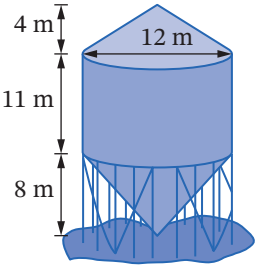
7 **a** Calculate the volume of this swimming pool. PS R



b Calculate the capacity of the pool if it is filled to a depth 20 cm from the top.

c If water costs \$1.98/kL, find the cost of filling the pool.

8 **a** Find the capacity of the silo shown, correct to the nearest m^3. PS R



b How many tonnes of barley grain will the silo hold if the density of barley grain is 600 kg/m^3?

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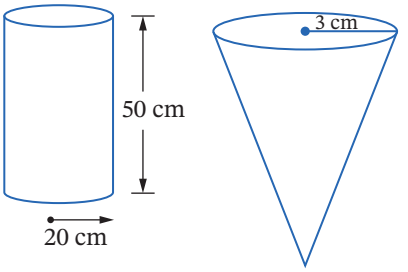
PATH

9 A drink container in the shape of a cylinder has a base radius of 20 cm and a height of 50 cm.

PS R

a Calculate the exact volume of the container.

b A full drink container can completely fill 800 cone-shaped cups with a radius of 3 cm. Calculate the height of each cup.



3.10 Areas of similar figures

PATH

Video
Areas of similar figures

Technology
Areas of similar shapes

Spreadsheet
Areas of similar shapes

Worksheet
Investigating paper sizes

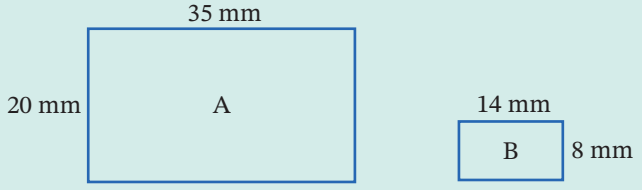
Areas of similar figures

- If the matching sides of 2 similar figures are in the ratio $1 : k$, then their areas are in the ratio $1 : k^2$.
- If the matching sides are in the ratio $m : n$, then their areas are in the ratio $m^2 : n^2$.

$$A_1 : A_2 = m^2 : n^2 \quad \text{or} \quad \frac{A_1}{A_2} = \frac{m^2}{n^2}$$

Example 25

What is the ratio of the areas of the similar rectangles shown?



Solution

Ratio of matching sides (A to B) $= 35 : 14$
$= 5 : 2$

Ratio of areas $= 5^2 : 2^2$
$= 25 : 4$

Example 26

2 similar pentagons have areas in the ratio 144 : 169. Find the ratio of the lengths of their matching sides.

Solution

Ratio of areas $= m^2 : n^2 = 144 : 169$

$\therefore$ Ratio of sides $= m : n = \sqrt{144} : \sqrt{169} = 12 : 13$

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Example 27

2 similar triangles have matching sides in the ratio 3 : 5. If the area of the larger triangle is 225 cm^2, find the area of the smaller triangle.

Solution

Let the area of the smaller figure be A.

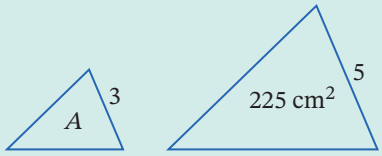
Ratio of matching sides = 3 : 5

Ratio of areas = $3^2 : 5^2 = 9 : 25$

$\therefore \frac{A}{225} = \frac{9}{25}$

$A = \frac{9}{25} \times 225 = 81$ cm^2

The area of the smaller triangle is 81 cm^2.



EXERCISE 3.10 ANSWERS ON P. 616

Areas of similar figures UFRC

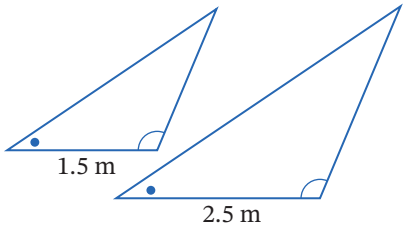
1 For each pair of similar figures, find the ratio of their areas.

EXAMPLE 25

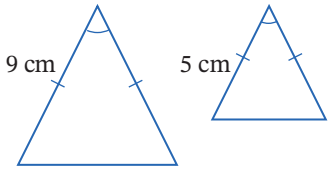
a



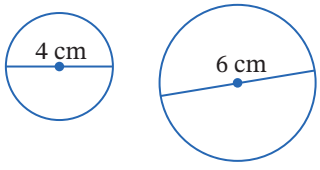
b



c



d



2 For each ratio of the areas of 2 similar figures, find the ratio of the lengths of their matching sides.

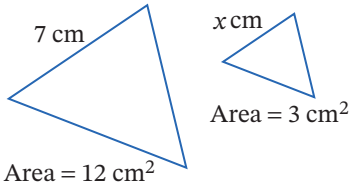
EXAMPLE 26

a 9 : 25 **b** 1 : 100 **c** 64 : 25 **d** 16 : 81

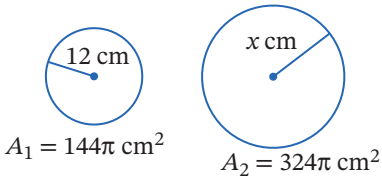
3 Find x for each pair of similar figures.

EXAMPLE 27

a



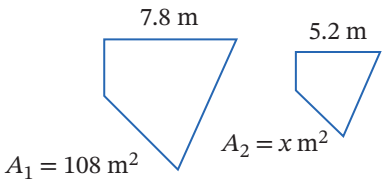
b



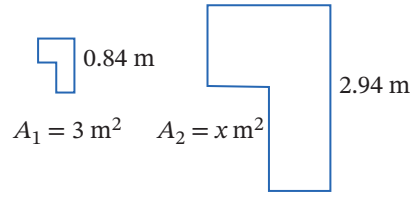
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PATH

c



d



4 2 circles have radii in the ratio 3 : 5. If the larger area is 150 cm^2, find the area of the smaller circle. R

5 2 squares have sides in the ratio 7 : 4. If the area of the smaller square is 14.4 cm^2, find the area of the larger square. R

6 2 similar triangles have areas in the ratio 4 : 9. If the length of the base of the smaller triangle is 5 cm, find the length of the base of the larger triangle. R

7 2 similar rectangles have their areas in the ratio 36 : 121. If the width of the smaller rectangle is 84 cm, find the width of the larger rectangle.

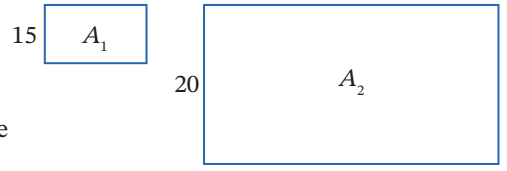
8 If the radius of a circle is doubled, how has its area changed? R C

9 If the area of a square is divided by 9, how has its side length changed? R C

10 If the sides of a triangle are increased by 2.5, how has its area changed? R C

11 If the area of a trapezium is decreased by $\frac{1}{100}$, how have its sides changed? R C

12 A_1 and A_2 are the areas of these 2 similar rectangles.



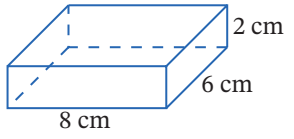
a Find $A_1 : A_2$.

b If the length and width of each rectangle is increased by 20%, find $A_1 : A_2$.

Investigation

Surface areas and volumes of similar solids

1 **a** Calculate the volume of this rectangular prism.



b Calculate the surface area of the rectangular prism.

c If the length, width and height are all doubled, what happens to:

i the volume? **ii** the surface area?

d Copy and complete:

If the length, width and height are all doubled, the volume is increased _____ times and the surface area is increased _____ times.

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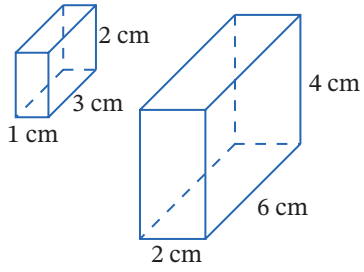
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2 **a** Explain why these rectangular prisms are similar solids.

b What is the ratio of their matching sides?

c What is the ratio of their surface areas?

d What is the ratio of their volumes?

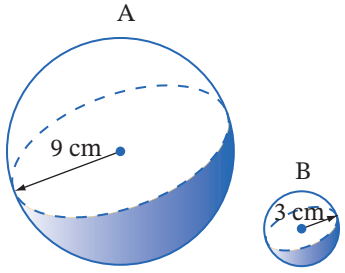


3 For the spheres A and B, find the ratio of:

a their radii

b their surface areas

c their volumes



4 How is the ratio of surface areas of similar solids related to the ratio of matching sides?

5 How is the ratio of volumes of similar solids related to the ratio of their matching sides?

PATH

3.11

Surface areas and volumes of similar solids 3.11

Surface areas and volumes of similar solids

- If the matching sides of 2 similar solids are in the ratio $1 : k$, then their surface areas are in the ratio $1 : k^2$ and their volumes are in the ratio $1 : k^3$.
- If the matching sides are in the ratio $m : n$, then their surface areas are in the ratio $m^2 : n^2$ and their volumes are in the ratio $m^3 : n^3$.

$$\frac{SA_1}{SA_2} = \frac{m^2}{n^2} \text{ and } \frac{V_1}{V_2} = \frac{m^3}{n^3}$$

PATH

Worksheet
Areas and volumes of similar figures

Videos
Surface areas and volumes of similar solids

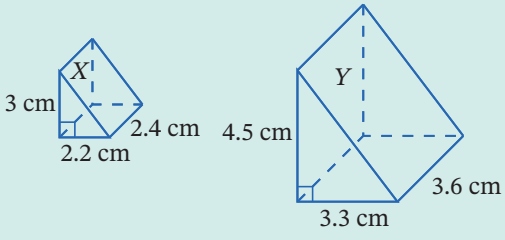
The incredible strength of ants

Example 28

For these 2 similar triangular prisms, find the ratio of their:

a surface areas

b volumes



PATH

Solution

a Ratio of sides (X to Y) $= 3 : 4.5$ (or $2.2 : 3.3$ or $2.4 : 3.6$)
$= 2 : 3$

Ratio of surface areas $= 2^2 : 3^2$
$= 4 : 9$

b Ratio of volumes $= 2^3 : 3^3$
$= 8 : 27$

Video
Surface areas and volumes of similar solids

Example 29

2 similar cylinders have their surface areas in the ratio 25 : 36. If the volume of the smaller cylinder is 250 cm^3, find the volume of the larger solid.

Solution

Ratio of surface areas $= 25 : 36$

$\therefore$ Ratio of matching sides $= \sqrt{25} : \sqrt{36}$
$= 5 : 6$

$\therefore$ Ratio of volumes $= 5^3 : 6^3$
$= 125 : 216$

Let the volume of the larger cylinder be V.

$$\frac{V}{250} = \frac{216}{125}$$
$$V = \frac{216}{125} \times 250$$
$$= 432$$

$\therefore$ Volume of the larger cylinder is 432 cm^3.

EXERCISE 3.11 ANSWERS ON P. 616

Surface areas and volumes of similar solids UFR

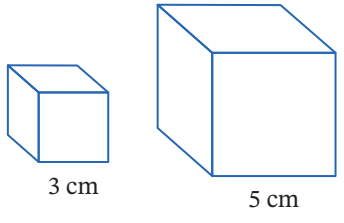
EXAMPLE 28

1 For each pair of similar solids, find the ratio of:

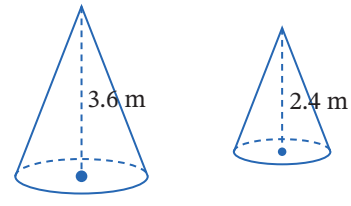
i the smaller surface area to larger surface area

ii the larger volume to the smaller volume

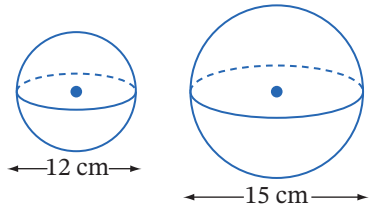
a



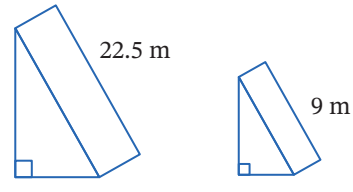
b



c



d



EXAMPLE 29

2 2 similar pyramids have surface areas of 81 cm^2 and 100 cm^2. Find the ratio of their:

a matching side lengths **b** volumes

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3 2 similar prisms have volumes of 125 cm^3 and 343 cm^3. Find the ratio of their:

a matching sides **b** surface areas

4 Blocks of chocolate are sold in the shape of similar triangular prisms. The areas of the triangular faces of two prisms are 6400 mm^2 and 1600 mm^2. If the volume of the smaller prism is 9600 mm^3, find the volume of the larger prism. R

5 2 drink cans have the shapes of similar cylinders. The larger can is 15 cm high and contains 350 mL of drink. If the smaller can is 9 cm high, how much drink does it contain? R

6 A box of washing powder is 12 cm tall and contains 750 g of washing powder. A similar box is 18 cm tall. How much washing powder does it contain? R

7 A large fish tank has a capacity of 624 L. A smaller similar fish tank has half the length, width and depth of the large tank. Find the capacity of the smaller tank. R

8 A cylinder has its height and radius increased 1.5 times. By what factor has its: R

a surface area increased? **b** volume increased?

9 A balloon has a radius of 8 cm. By what factor is the volume decreased if the radius changes to 6 cm? R

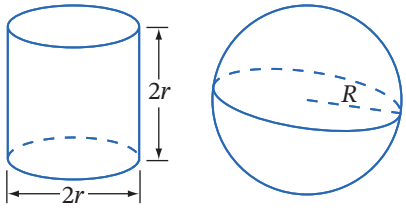
Power plus ANSWERS ON P. 617

1 A square prism and square pyramid have the same base and the same surface area.

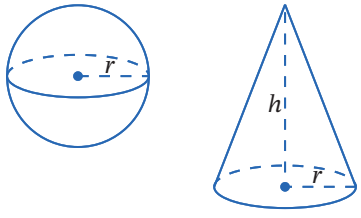
Show that the slant height, l, of the pyramid is $l=\frac{5}{2}s$, where s is the length of the base.

2 A cylinder with diameter and height $2r$ has the same surface area as a sphere of radius R.

Show that $R=\sqrt{\frac{3}{2}}r$.



3 A sphere and a cone have the same radius and volume. Show that the cone's height, h, is 4 times the radius, r.



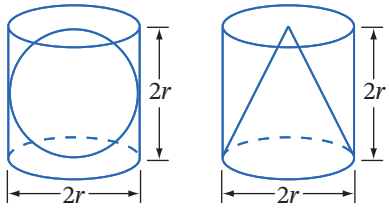
4 A sphere and a cone fit inside identical cylinders with the same base diameter and height.

a Find the ratio

Volume of cone: Volume of sphere : Volume of cylinder.

b Show that

Volume of cone + Volume of sphere = Volume of cylinder.



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CHAPTER 3 REVIEW

Quiz
Language of maths 3

Puzzle
Surface area and volume crossword (Advanced)

Language of maths

absolute error	apex	capacity	cone
cross-section	curved surface	cylinder	diameter
hemisphere	percentage error	perpendicular height	precision
pyramid	radius	ratio	rounding error
sector	similar figures	similar solids	slant height
sphere	surface area	truncate	volume

1 What is the difference between rounding and truncating a number?

2 Name 3 solids that have a **curved surface area**.

3 What is the formula for the curved surface area of a cone?

4 Explain the difference between the **perpendicular height** and the **slant height** of a pyramid.

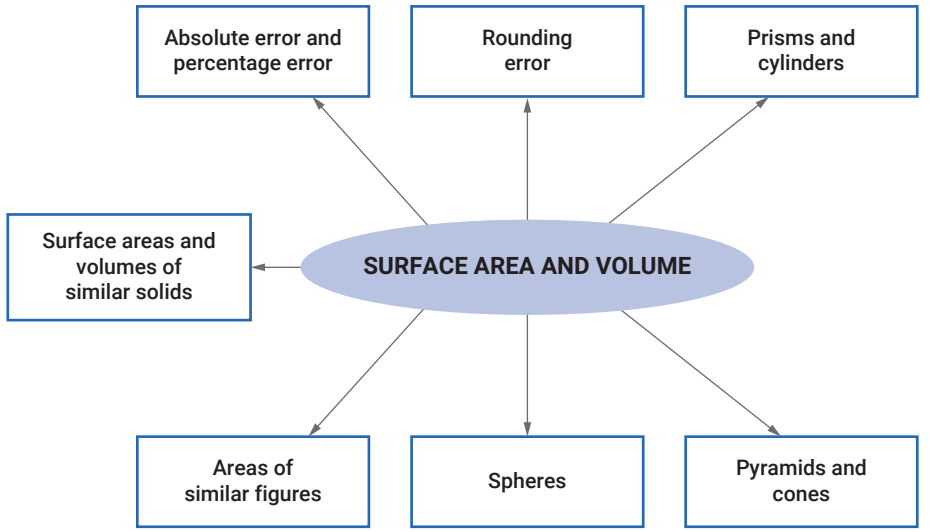
5 What is the formula $V = \frac{1}{3}\pi r^2 h$ used for?

6 Describe the relationship between the volumes of similar solids.

Topic summary

Worksheet
Mind map: Surface area and volume (Advanced)

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

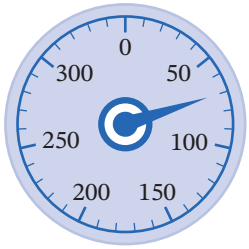


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TEST YOURSELF 3

ANSWERS ON P. 617

1 This weight scale shows a measurement of 70 g. Find the measurement's: 3.01



a absolute error

b percentage error (correct to one decimal place)

2 A plant stem is measured to be 16.0 cm correct to 1 decimal place. What is the percentage error in this measurement? 3.01

A 0.3125 % **B** 0.625 % **C** 3.125 % **D** 6.25 %

3 A triangle has a base length of 5.65 cm and a perpendicular height of 11.31 cm. 3.02

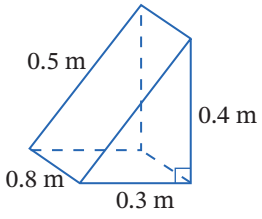
a Calculate its area.

b Round the triangle's base and height to one decimal place, use these values to calculate its area, then find correct to one decimal place the percentage error of the calculated area.

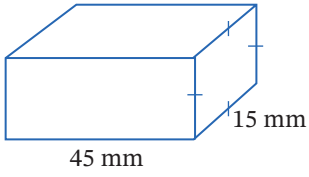
c Truncate the triangle's base and height to the nearest whole number, use these values to calculate its area, then find correct to one decimal place the percentage error of the area.

4 Find the surface area of each prism. 3.03

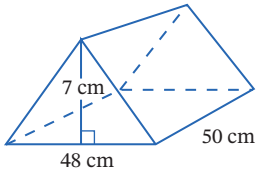
a



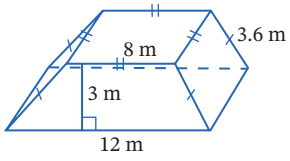
b



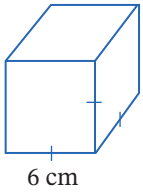
c



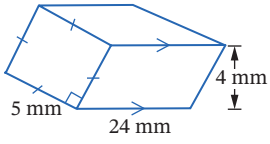
d



e



f



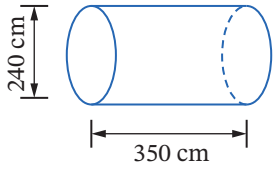
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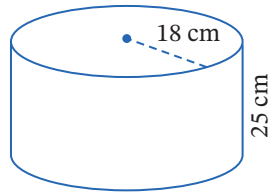
3.03

5 Calculate, correct to one decimal place, the surface area of each solid.

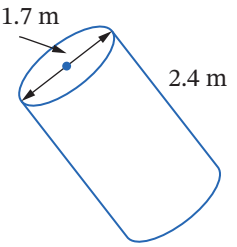
a



b

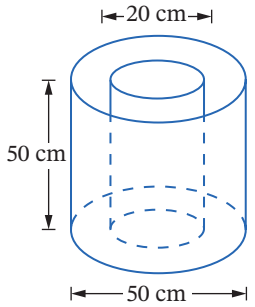


c

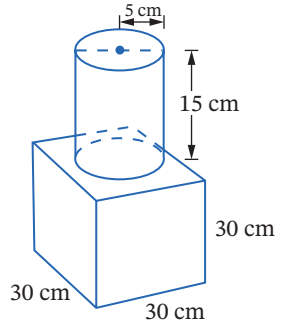


Cylinder, open at one end

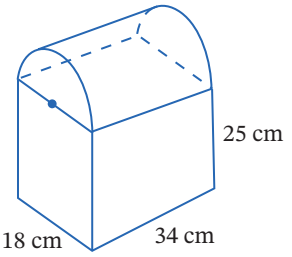
d



e



f

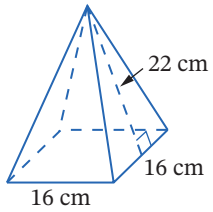


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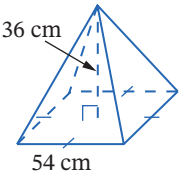
3.04

6 Find the surface area of each pyramid.

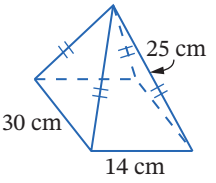
a



b



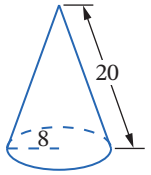
c



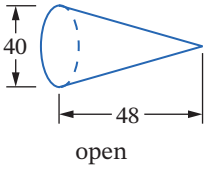
3.05

7 Find the surface area of each solid. All measurements are in metres.

a

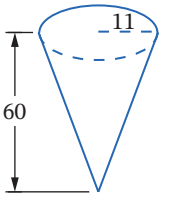


b

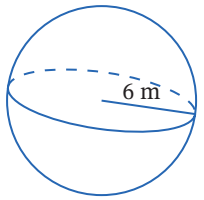


open

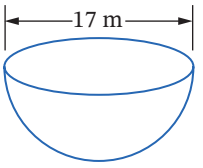
c



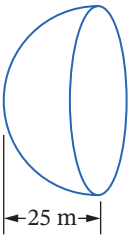
d



e



f



Foundation Standard Complex

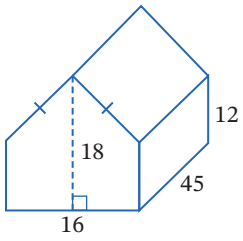
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PATH

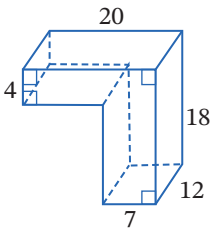
3.06

8 Find, correct to the nearest square centimetre, the surface area of each solid. All measurements are in centimetres.

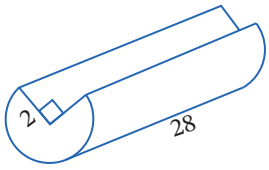
a



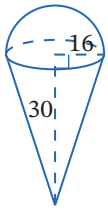
b



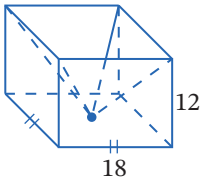
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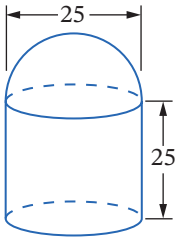
d



e



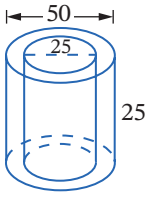
f



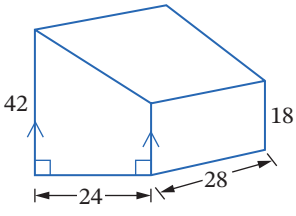
3.07

9 Calculate, correct to nearest cubic metre, the volume of each solid. All measurements are in metres.

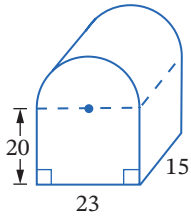
a



b



c

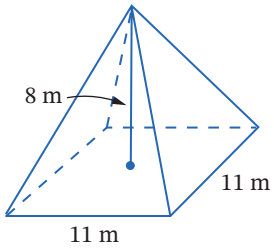


PATH

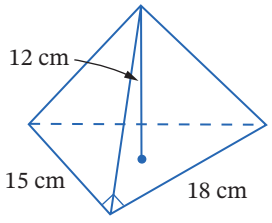
3.08

10 Find, correct to 2 decimal places (where necessary), the volume of each solid.

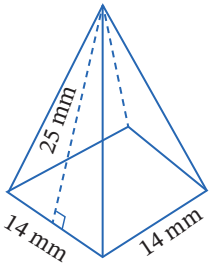
a



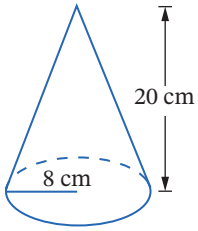
b



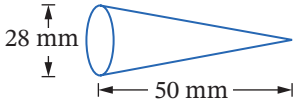
c



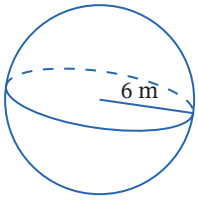
d



e



f



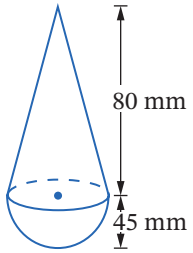
Foundation | Standard | Complex

PATH

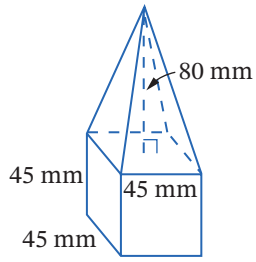
3.09

11 Find, correct to the nearest whole number, the volume of each solid.

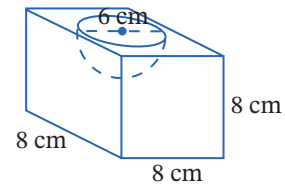
a



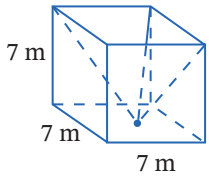
b



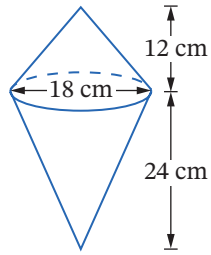
c



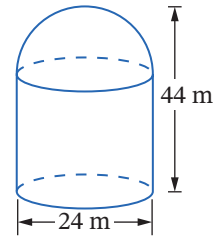
d



e



f



3.10

12 **a** 2 circles have radii in the ratio 4 : 5. If the smaller area is 150 cm^2, find the area of the larger circle.

b The radius of a circle is increased by a factor of $2\frac{1}{2}$. By what factor has the area increased?

3.11

13 **a** The areas of the bases of 2 similar rectangular prisms are in the ratio 25 : 64. If the volume of the larger prism is 1024 cm^2, find the volume of the smaller prism.

b 2 similar pyramids have volumes of 216 cm^3 and 343 cm^3. Find the ratio of their surface areas.

Foundation Standard Complex

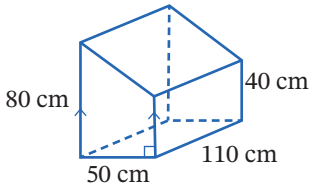
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PRACTICE SET 1

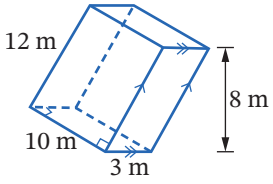
ANSWERS ON P. 617

1 Find the surface area of each prism. 3.03

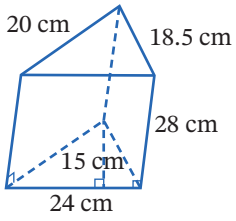
a



b



c



2 Find, correct to the nearest degree, the angle of inclination of a line with gradient $-\frac{3}{4}$. PATH 2.01

3 Naved is paid 4 weeks' normal pay plus a 17.5% annual leave loading for his 4-week holiday. If Naved's salary is $78 580, find his: 1.01

a normal weekly pay **b** leave loading

c total pay for the 4-week holiday.

4 A call centre operator is employed under the award shown in the table. 1.01

Normal rate: $28.75 per hour	
Normal rate	For 0 to 36 hours worked
Time-and-a-half	For the next 4 hours worked
Double time	For each hour worked after that

Calculate the wage for working:

a 20 hours **b** 39 hours

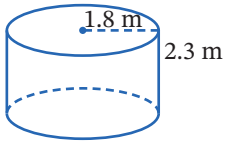
c 45 hours

5 A line passes through the points $H(5, -3)$ and $K(8, -7)$. Calculate the gradient of the line: 2.02

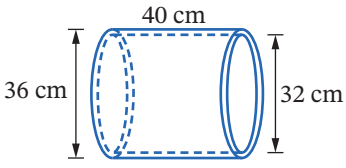
a parallel to HK **b** perpendicular to HK

6 Calculate, correct to one decimal place, the surface area of each solid. 3.03

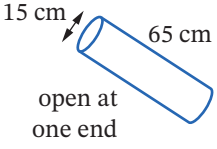
a



b



c



7 Measuring with a ruler, Timothee gives a measurement as rounded to 26 cm. 3.01

a What is the precision of this ruler?

b What is the absolute error of Timothee's measurement?

c Using a ruler marked in millimetres, Lionel gives the same measurement as 26.8 cm. What is the absolute error of this measurement?

d Calculate correct to 2 decimal places the percentage error of Timothee's and Lionel's measurements.

e Whose measurement is more accurate, Timothee's or Lionel's?

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1.02 **8** Weng earns a weekly wage of $1485. She has annual deductions of $1756 for a health fund and $3560 for work expenses.

a Calculate Weng's taxable income.

b Use the tax table on page 11 to calculate the income tax Weng should pay.

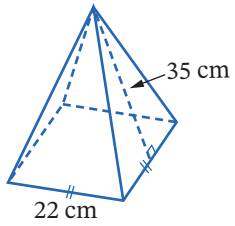
2.03 **9** Graph the linear equations $y = 3x - 2$ and $y = -2x + 3$ on a number plane. Where do the lines intersect?

PATH

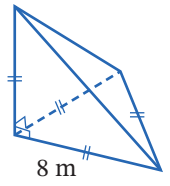
10 Find the surface area of each pyramid.

3.04

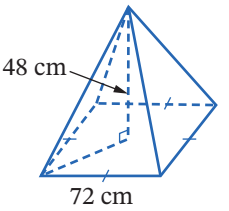
a



b

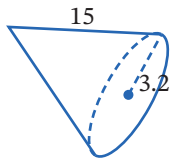


c

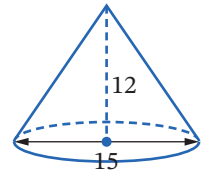


3.05 **11** Find, correct to the nearest square metre, the surface area of each solid. All measurements are metres.

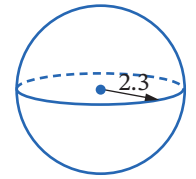
a



b



c



2.05 **12** Find the equation of the line passing through the point $(-8, 3)$ with gradient $-\frac{2}{3}$.

2.03 **13** Which of the following points lie on the line of $2x + y = 3$?

Select the correct answer **A**, **B**, **C** or **D**.

A $(1, 0)$ **B** $(2, -1)$ **C** $(-1, -1)$ **D** $(-1, -5)$

1.05 **14** Calculate the value of an investment if $46 000 is invested at 4.2% p.a. for 3 years with interest compounded:

a annually **b** quarterly **c** monthly

PATH

2.04 **15** Write the linear equation $y = \frac{3}{4}x + 6$ in general form $ax + by + c = 0$.

2.04 **16** Find the gradient and y-intercept of the line with equation $5x - 2y + 3 = 0$.

1.06 **17** Jovan purchases a home theatre system valued at $7680. He pays a 10% deposit and repays the balance in 48 monthly instalments. Interest on the balance is charged at a flat rate of 14.5% p.a. Find:

a the deposit paid **b** the balance owing

c the interest charged **d** the total amount to be repaid

e the amount of each instalment **f** the total price Jovan paid for the system

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18 Annieke purchases a new car for $39 990, which depreciates by 10% p.a. 1.07

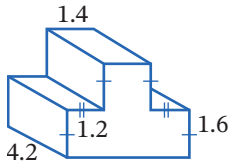
a Find the salvage value of the car after 4 years.

b What is the depreciation over this time?

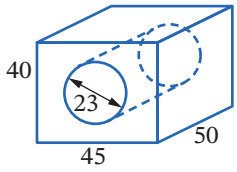
c Express the salvage value as a percentage of the cost price (to one decimal place).

19 Calculate, to nearest cubic metre, the volume of each solid. All measurements are in metres. 3.09

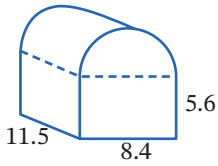
a



b



c



20 The interest earned on $8500 over 3 years is $624.75. Calculate the flat rate of interest p.a. 1.03

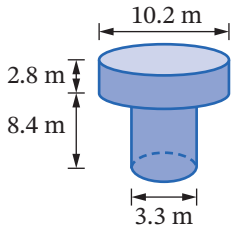
21 Find, in general form, the equation of a line which passes through the points $(3, -2)$ and $(-1, 3)$. PATH

22 The line $2x + 5y + 7 = 0$ and another line intersect at right angles at the point $(4, -5)$. 2.05

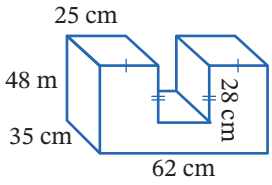
Find the equation of the other line.

23 Find the surface area of each solid, correct to 1 decimal place. 2.08 3.06

a



b



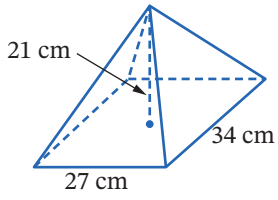
24 Rewrite each equation in the form $y = mx + c$. 2.04

a $5x - y + 3 = 0$ **b** $4x + y - 8 = 0$ **c** $2x - 6y + 8 = 0$

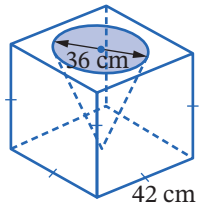
25 Show that the points $P(-2, -1)$, $Q(3, 2)$, $R(0, 7)$ and $S(-5, 4)$ are the vertices of a square. 2.09

26 Find, correct to one decimal place, the volume of each solid. 3.09

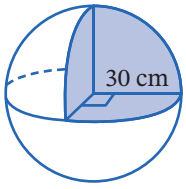
a



b



c



27 2 similar squares have sides in the ratio 3 : 5. If the smaller area is 360 cm^2, find the area of the larger square. 3.10

28 2 similar containers have volumes 600 cm^3 and 2025 cm^3 respectively. If the height of the larger container is 60 cm, what is the height of the smaller container? 3.11

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